

Structure–Activity Relationship of Antibody–Oligonucleotide Conjugates: Evaluating Bioconjugation Strategies for Antibody–siRNA Conjugates for Drug Development

Michael Cochran, Danny Arias, Rob Burke, David Chu, Gulin Erdogan, Michael Hood, Philip Kovach, Hae Won Kwon, Yanling Chen, Michael Moon, Christopher D. Miller, Hanhua Huang, Arthur Levin, and Venkata Ramana Doppalapudi*



Cite This: *J. Med. Chem.* 2024, 67, 14852–14867



Read Online

ACCESS |



Metrics & More



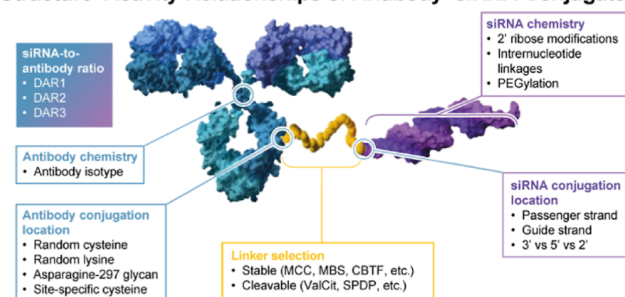
Article Recommendations



Supporting Information

ABSTRACT: Antibody–oligonucleotide conjugates are a promising class of therapeutics for extrahepatic delivery of small interfering ribonucleic acids (siRNAs). These conjugates can be optimized for improved delivery and mRNA knockdown (KD) through understanding of structure–activity relationships. In this study, we systematically examined factors including antibody isotype, siRNA chemistry, linkers, conjugation chemistry, PEGylation, and drug-to-antibody ratios (DARs) for their impact on bioconjugation, pharmacokinetics (PK), siRNA delivery, and bioactivity. Conjugation site (cysteine, lysine, and Asn297 glycan) and DAR proved critical for optimal conjugate PK and siRNA delivery. siRNA chemistry including 2' sugar modifications and positioning of phosphorothioates were found to be critical for delivery and duration of action. By utilizing cleavable and noncleavable linkers, we demonstrated the impact of linkers on PK and mRNA KD. To achieve optimal properties of antibody–siRNA conjugates, a careful selection of siRNA chemistry, DAR, conjugation sites, linkers, and antibody isotype is necessary.

Structure–Activity Relationships of Antibody–siRNA Conjugates



INTRODUCTION

Antibodies and oligonucleotides, two powerful classes of molecules, have diverse applications in biology and medicine. As highly specific proteins, antibodies recognize and bind to a wide range of targets. Their proven effectiveness as receptor antagonists and as delivery vehicles for small molecules makes them potent therapeutics.^{1–4}

Oligonucleotides exert their therapeutic effect through multiple mechanisms, including inducing the degradation of mRNA or promoting splice modulation.^{5–9} Over the last 25 years, more than 16 oligonucleotide therapeutics have been approved for multiple therapeutic indications, including homozygous familial hypercholesterolemia, spinal muscular atrophy, Duchenne muscular dystrophy, hereditary transthyretin-mediated amyloidosis, familial chylomicronemia syndrome, acute hepatic porphyria, and primary hyperoxaluria.¹⁰

Small interfering ribonucleic acids (siRNAs) are 20–25 nucleotide-long, double-stranded RNA molecules containing a complementary sense and antisense strand. When inside the cell, these duplexes load into the Ago2 protein. Following this, the sense strand of the duplex siRNA is cleaved, and its fragments ejected, leaving behind a functional RNA-induced silencing complex (RISC) that comprises the antisense strand and Ago2. The RISC then uses the antisense strand to

recognize and bind to complementary mRNA sequences within the cell. Once fully hybridized with the mRNA, Ago2 cleaves the mRNA between target bases 10 and 11, resulting in mRNA degradation and reduced gene expression. siRNAs are proven therapeutic agents, especially in diseases caused by the overexpression of certain genes or the presence of harmful genetic mutations. By selectively targeting specific disease-related RNA, siRNA-based therapies hold promise for treating a wide range of conditions, including certain types of cancer, viral infections, and genetic disorders.^{5–9}

siRNA therapeutics face three main challenges: selectivity, stability, and delivery. Selectivity is primarily tackled through bioinformatics or chemical modifications, which can help predict the efficacy of candidate oligonucleotides and ensure they are designed to minimize off-target effects.¹¹ Stability is addressed by chemical modifications, such as phosphorothioate linkages or 2'-O-methyl modifications, which protect

Received: April 18, 2024

Revised: July 30, 2024

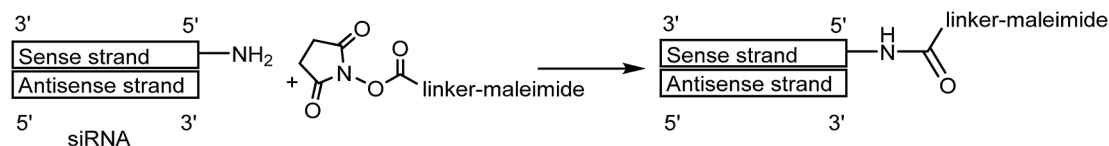
Accepted: August 5, 2024

Published: August 28, 2024

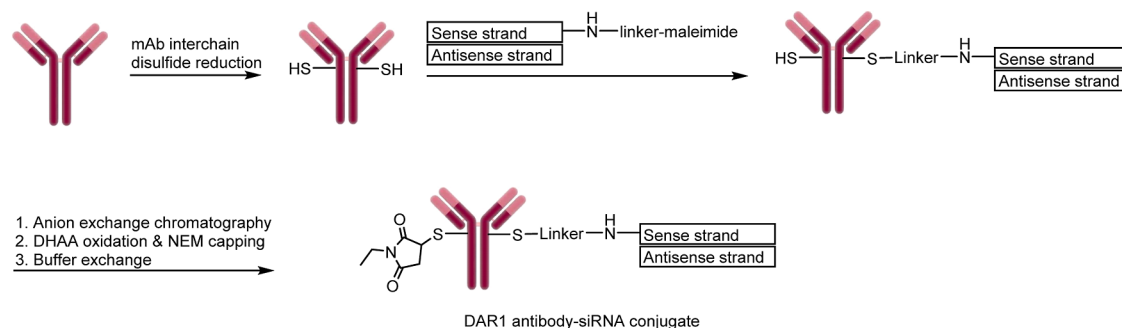


Scheme 1. (a) Synthetic Scheme for Conjugating an siRNA with a Linker; (b) Antibody–siRNA Conjugate Synthesis via Interchain Disulfide (Cysteine) Conjugation; (c) Antibody–siRNA Conjugate Synthesis via Asn297 Conjugation; (d) Antibody–siRNA Conjugate Synthesis via Lysine Conjugation

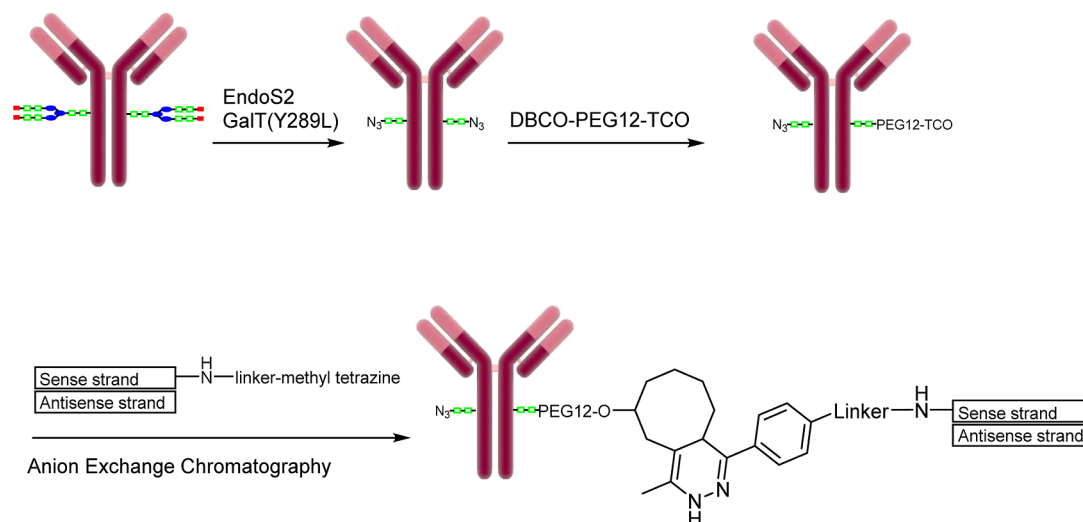
a.



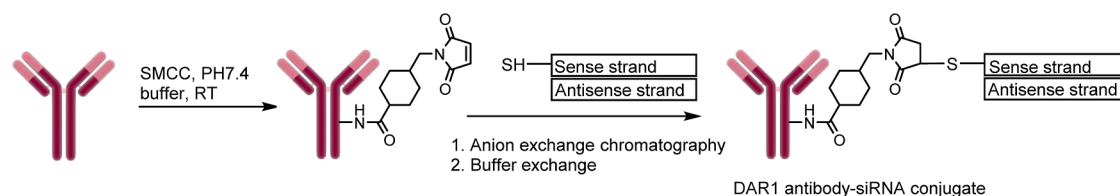
b.



c.



d.



the oligonucleotides from degradation by nucleases in the bloodstream and other tissues. Delivery is an issue because different tissue types can present unique barriers, requiring specific targeted delivery strategies dependent on the target tissue. siRNA delivery to the liver is achieved with lipid nanoparticle formulations and conjugation of ligands such as

N-acetylgalactosamine (GalNAc), which has a high affinity for asialoglycoprotein receptors (ASGPRs) that are predominantly expressed on hepatic parenchymal cells. Also, intrathecal delivery of siRNA modified with C16, or 2'-*O*-hexadecyl has produced potent and durable knockdown (KD) in the central nervous system (CNS).¹² However, there remains a significant

unmet need for technologies that can effectively deliver oligonucleotides systemically to other tissues.¹³ Owing to their inherent specificity for cell surface receptors and their demonstrated capability to transport small molecules to a variety of tissues, antibodies have emerged as promising vehicles for oligonucleotide delivery. Antibody–oligonucleotide conjugates (AOCs) present a promising solution for overcoming the challenges associated with oligonucleotide delivery.

Previous research by different groups has explored the use of antibody-mediated targeted delivery of oligonucleotides, yielding varying degrees of success. Seminal work by Cuellar et al. (2015) showed that siRNA oligonucleotides could be conjugated to thiol-containing monoclonal antibodies (mAbs), resulting in active, stable conjugates in vitro.¹⁴ They showed that the mAb–siRNA conjugates retained a similar binding affinity and gene-silencing capacity to that of their constituent mAb or siRNA. However, when tested in mouse tumor xenograft studies, their activity was modest. Sela et al. (2023) recently reported on a new method for delivering antisense-oligonucleotides (ASOs) to the CNS for treating neurological disorders.¹⁵ They conjugated the ASO to an antibody that could cross the blood–brain barrier. The study found promising in vitro activity and in vivo pharmacokinetics (PK) behavior, suggesting potential for future ASO-based CNS drugs. Of particular interest is the targeted delivery of various oligonucleotides to muscle for the treatment of muscle disorders via a transferrin receptor 1 (TfR1)-mediated approach.¹⁶ Recent publications, such as the work by Malecova et al. (2023), have demonstrated effective delivery of diverse oligonucleotides to muscle tissue.¹⁷

While other studies have demonstrated delivery of oligonucleotides conjugated to mAbs and antigen-binding fragments, there is currently a lack of information on comprehensive SARs of antibody–siRNA conjugates.^{16,18–20} This publication will provide a comprehensive SAR of antibody–siRNA conjugates, including optimal linker selection, points of conjugation on the antibody and siRNA, bioconjugation strategies to link an antibody and siRNA oligonucleotide, and the impact of siRNA oligonucleotide-to-antibody ratios on activity. The AOC approach offers a flexible strategy to increase the potential delivery of oligonucleotides to target tissues.

■ AOC NAMING CONVENTION

In the rapidly evolving field of AOCs, the need for a clear, standardized naming convention is paramount because a multitude of variables exist, such as the type of antibody, its target, the linker used, and the siRNA conjugated. Our naming convention, detailed below, offers a systematic approach using specific symbols and abbreviations to denote each component of an AOC.

Take the following conjugates as examples: *α*hEGFR-Cys-MCC-siKRAS, *α*mTfR1-Cys-MCC-siMstn, *α*mTfR1-Cys-PDP-siMstn (for disulfide linkers). Here, “*α*” means “anti” to indicate that the antibody binds to EGFR (epidermal growth factor receptor) or TfR1. “h” or “m” denote whether the antibody is human (h) or mouse (m) specific. “Cys” or “Lys” represent the amino acids cysteine (Cys) or lysine (Lys) that the linker is attached to on the antibody. “MCC” or “PDP” are the linkers, 4-(*N*-maleimidomethyl)cyclohexane-1-carboxamide and 3-(2-Pyridyldithio)propionamide. The “siKRAS” or “siMstn” are the siRNAs, targeting KRAS mRNA and Mstn

mRNA, respectively, that are conjugated to the antibody. For example, “*α*hEGFR-Cys-MCC-siKRAS” would be an anti-EGFR antibody that is specific to humans, with an MCC linker attached to a cysteine residue, and conjugated to an siRNA targeting KRAS.

In our naming convention for linkers, we use the linker name, rather than the reagent used in the reaction. This is because during the reaction, a part of the reagent is removed. For instance, instead of using “SMCC” (the reagent) as the name of the linker, we simply use “MCC”. This approach ensures that the names we use are representative of the actual structure of the AOCs. However, in the [Experimental Section](#), we use the full reagent name when discussing the linker reaction with the siRNA; for example, SMCC (reagent) versus MCC (linker).

■ RESULTS AND DISCUSSION

We first describe a general overview of the AOC synthesis used to generate antibody–siRNA conjugates. AOCs can be generated using any antibody isotype and siRNA with a conjugation handle, using cysteine, lysine or Asn297 conjugation methods and their corresponding linkers. The conditions and yields may vary, based on the constructs being used, but the general scheme remains the same.

We then detail the results of bioconjugation reaction and purification steps that reduce the risk of product aggregation and yield a highly pure AOC product. We then describe our investigations into antibody–siRNA conjugate SARs. We evaluated siRNA chemical modifications, siRNA conjugation locations on the mAb, cleavable versus noncleavable linkers, linker location on the siRNA, the addition of polyethylene glycol (PEG), and antibody isotypes.

[Table S1](#) describes the antibody target, linker, siRNA sequence and chemical modification pattern for each antibody–siRNA conjugate. Furthermore, [Table S2](#) provides a comprehensive list of the siRNA conjugation components, detailing the siRNA conjugation handles, linker reagent structures, linker–siRNA structures, antibody reactive groups, and final antibody–siRNA conjugate structures.

AOC Synthesis. General Synthetic Scheme for Antibody–siRNA Conjugate Generation. [Scheme 1](#) outlines the general synthetic scheme we use to generate antibody–siRNA conjugates. For cysteine-conjugated AOCs, we first couple a linker–maleimide to the amino group at the 5′ end of the sense strand of a duplex siRNA using *N*-hydroxysuccinimide (NHS) chemistry.^{21,22} This generates an intermediate siRNA–linker–maleimide ([Scheme 1a](#)).

We then partially reduce the interchain disulfide of the antibody with tris(2-carboxyethyl)phosphine (TCEP) ([Scheme 1b](#)). This means the disulfide bonds between the cysteine residues in the antibody are broken, exposing the cysteine residue thiol groups, which can then be used for conjugation with the oligonucleotide. The antibody is then conjugated to the siRNA–linker–maleimide. The resulting intermediate conjugate is processed through anion exchange chromatography, dehydroascorbic acid (DHAA) oxidation, *N*-ethylmaleimide (NEM) capping, and buffer exchange. This results in an antibody–siRNA conjugate with a drug-to-antibody ratio of 1 (DAR1). [Scheme 1b](#) depicts random cysteine conjugation. In this paper, we will also discuss site-specific cysteine conjugation.

In addition to the cysteine conjugation shown in [Scheme 1b](#), AOCs can be synthesized via Asn297 ([Scheme 1c](#)) and lysine

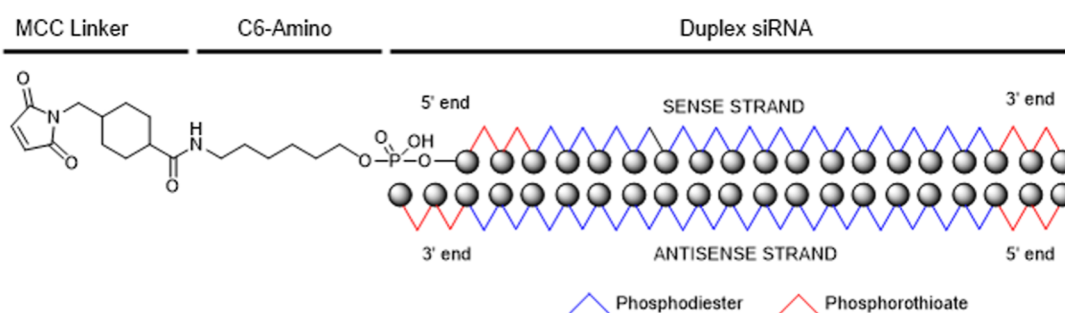


Figure 1. Typical structure of an siRNA molecule. MCC linker: a heterobifunctional cross-linker that contains two reactive groups: an NHS ester and a maleimide. The NHS ester reacts with primary amines, while the maleimide reacts with sulfhydryl groups. This allows the MCC to form stable bonds with both amine- and sulfhydryl-containing molecules. C6 amino: a linker between the MCC and the siRNA. It provides a sufficient distance between the siRNA and the MCC to minimize any potential steric hindrance or interference with the siRNA's function. Sense strand 5' end: the 5' end of the sense strand of the siRNA. The 5' end is where the MCC is attached via the C6 amine linker. Sense strand 3' end: the 3' end of the sense strand of the siRNA. Antisense strand: designed to be complementary to the target mRNA sequence. It guides the RISC to the target mRNA. Phosphodiester and phosphorothioate bonds: the bonds that connect the nucleotides in the siRNA. Phosphodiester bonds are the standard bonds in RNA, while phosphorothioate bonds are modified versions that contain a sulfur atom, which can provide increased stability to the siRNA.

(Scheme 1d) conjugations.^{23–25} For synthesis via Asn297, we modify the N-linked glycans at Asn297 enzymatically to azide-activate the antibody on each heavy chain. A bifunctional dibenzocyclooctyne (DBCO)-PEG-trans-cyclooctene (TCO) linker is added to the azide conjugation handle on the antibody to generate an antibody with a TCO conjugation handle at the Asn297 location. In a separate reaction, methyltetrazine-siRNA is generated by conjugating methyltetrazine NHS ester to the siRNA amine conjugation handle. A TCO-methyltetrazine click chemistry is performed between the functionalized antibody and siRNA to generate an AOC with a DAR of 1. For synthesis via lysine, the antibody is activated using the noncleavable linker succinimidyl-4-(*N*-maleimidomethyl)-cyclohexane-1-carboxylate (SMCC); the NHS ester of SMCC reacts with the amino groups of the lysine residues attaching reactive maleimide groups to the lysine side chain. The maleimide-activated antibody is then reacted with the thiol-containing siRNA and the resulting intermediate conjugate is then processed through anion exchange chromatography and buffer exchange, resulting in a DAR1 antibody-siRNA conjugate.

Figure 1 shows a typical structure of an siRNA molecule conjugated with a 4-(*N*-maleimidomethyl)cyclohexane-1-carboxamide (MCC) linker through a C6 amine linker. This paper describes AOCs generated with siRNAs conjugated at the 5' and 3' termini of the sense strand. There are potential advantages of each of these strategies, including using the steric bulk of the conjugation to protect against 3' nucleases as well as preventing unwanted loading of the sense strand into RISC by conjugating to the 5' end.²⁶

Throughout this paper, AOC dose levels are based on the siRNA component of the AOC. Calculations to convert siRNA to mAb or AOC concentrations are provided in the Experimental Section. A 1 mg/kg siRNA dose is equivalent to a 10.6 mg/kg mAb dose, and an 11.6 mg/kg DAR1 AOC dose. While the molecular weight of the siRNA and AOC may change with sequence, modifications, and linkers, the impact of these changes is less than 5% of the total MW and does not significantly impact these ratios.

Optimization of Disulfide Bond Reduction and Capping in Cys Conjugation Method. The cysteine DAR 1 AOCs generated by partially reducing the interchain disulfide bonds of an antibody have 1–7 unreacted thiols depending upon the

number of reduced disulfides in an antibody molecule. These free cysteines could lead to aggregates, disulfide scrambling, and impact the conjugate stability due to a fewer number of disulfide bonds holding the conjugate together. To preemptively solve these potential challenges, we used oxidation with DHAA to reform the reduced excess disulfide bonds, then a capping of the remaining one cysteine of the isolated DAR1 AOC using NEM alkylation to form a thioether.

We evaluated the number of free cysteines on a DAR1 conjugate (Figure 2). After TCEP reduction, siRNA

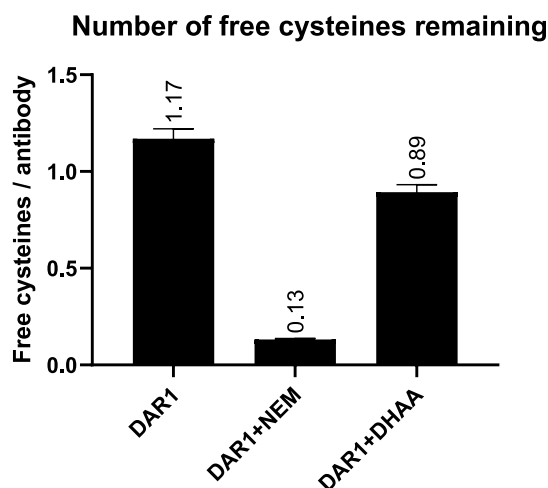


Figure 2. Evaluation of free cysteines available for conjugation on an α hEGFR-Cys-MCC-siAR DAR1 AOC following treatments with TCEP, NEM, and DHAA ($n = 3$).

conjugation, and purification using strong anion exchange chromatography (SAX), DAR1 α hEGFR-siRNA AOCs (α hEGFR-Cys-MCC-siAR) without DHAA oxidation or NEM capping had more than one free cysteine available, indicating that some DAR1 conjugate molecules contain at least one other reduced disulfide bond. This isolated DAR1 AOC was split into three treatment groups: no treatment (DAR1), five equivalents of NEM to cap unreacted cysteine (DAR1 + NEM), and 10 equiv of oxidizing DHAA (DAR1 + DHAA). NEM capping blocked nearly all free cysteines (0.13 free cysteines per antibody) of DAR1 AOC, while DHAA

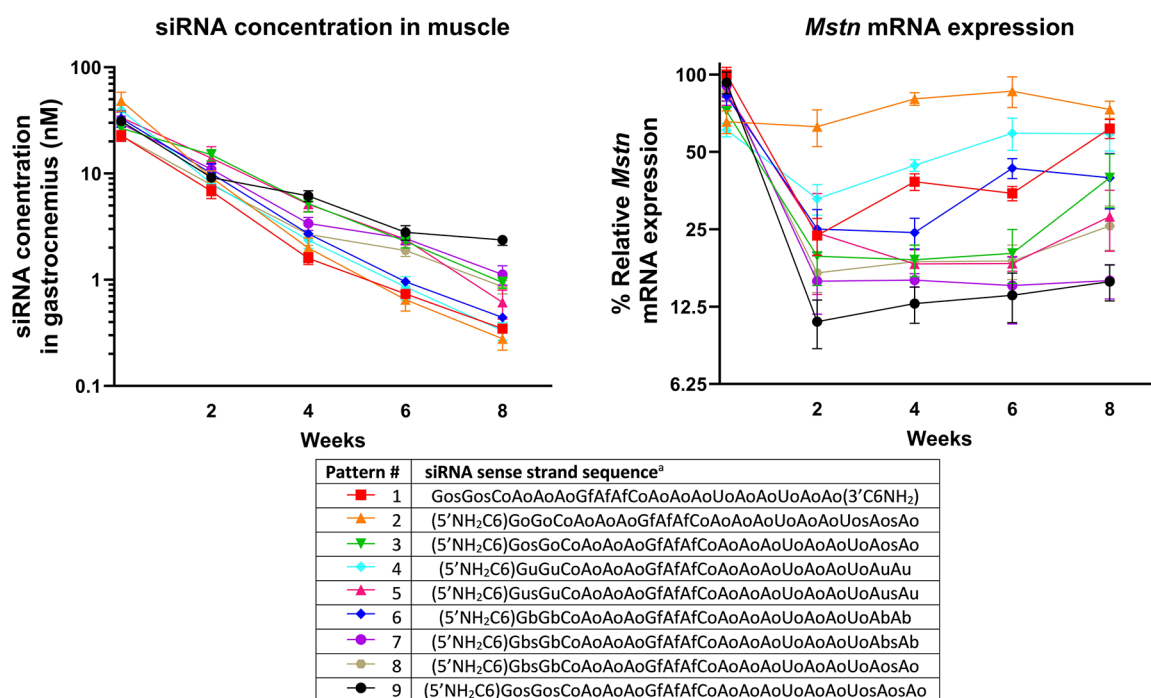


Figure 3. Impact of siRNA chemical modifications on siRNA stability and activity in targeting *Mstn* through evaluation of tissue siRNA concentration (TC) and percent mRNA remaining relative to phosphate buffered saline (PBS) in gastrocnemius muscle of mice treated with α mTfR1-Cys-MCC-si*Mstn* AOCs. ^aAll sense strands were duplexed with the antisense strand: vpUusUfsAoUoUoAfUoUoGoUoUoCoUfUoUfGoCoCosUosUo. Abbreviations: X refers to bases evaluated (AUGC): Xo = 2'-O-methyl, Xf = 2'-Fluoro, Xb = LNA, Xu = UNA-2'MOE, s = phosphorothioate, vp = vinylphosphonate.

oxidation left one free cysteine available for potential conjugation (0.89 free cysteines per antibody). AOCs with a DAR of 1 were at least 95% pure by strong anion exchange chromatography and size exclusion chromatography (SEC). The results suggest that after oxidation with DHAA, the α hEGFR hIgG1 DAR1 AOC used in this experiment will have approximately one free cysteine available to conjugate other functional groups (such as peptides or fluorescent dyes) or that free cysteine must be capped by alkylation to prevent AOC dimer formation in storage conditions (data not shown). These results are specific to the antibody used in this experiment. It is expected that each antibody will have its own degree of reduction after TCEP treatment. However, the presence of free cysteines and the requirement for alkylation to reduce dimer formation have been observed for DAR1 AOCs made with two other hIgG1 antibodies (data not shown).

siRNA Chemical Modifications. The susceptibility of the siRNA internucleotide phosphodiester bond to degradation by endonucleases or exonucleases upon systemic administration hinders the accumulation of intact therapeutic siRNA in target tissues. It is important for siRNAs to be stable in circulation as well as in tissues to show not only initial target mRNA KD, but also long duration of action after deposition in the target tissues. Chemically modifying siRNAs, such as replacement of 2'-OH with a 2'-deoxy-2'-fluoro (2'-F), 2'-O-methyl (2'-OMe), or 2'-O-methoxyethyl (2'-MOE) group, as well as substitutions of certain internucleotide phosphates with phosphorothioates (P=S) is a widely used strategy to improve siRNA stability in circulation as well as intracellular environments.^{27,28} One commonly used modification pattern includes two P=S at each terminus of the antisense strand, two P=S at the 5' end of the sense strand, and either one or no P=S on the 3' end of the sense strand, which is frequently used for

GalNAc conjugates with a 3'-pyrrolidine linker.^{10,29} We tested the same pattern with a C6 amine linker as an AOC and found it to be either less active or to have low durability with AOCs in vivo, as shown in Figure 3. We presume this is due to nuclease-mediated cleavage of siRNA from the unprotected end of the sense strand in mice.

To improve the stability and activity of siRNAs on AOCs in muscle tissue, we explored combinations of sugar modifications including locked nucleic acid (LNA) and unlocked nucleic acid (UNA), which are known to improve siRNA stability in biological matrices, and phosphorothioate patterns. We used AOCs consisting of an antimouse TfR1 mAb (α mTfR1) conjugated to siRNAs targeting the myostatin (*Mstn*) gene with the MCC linker (α mTfR1-Cys-MCC-si*Mstn*). The modification pattern of each sense strand is shown in Figure 3. Each sense strand was duplexed with the same antisense strand (vpUusUfsAoUoUoAfUoUoGoUoUoCoUfUoUfGoCoCosUosUo) containing an unlocked 2'-O-methoxymethyl uridine base with a vinylphosphonate at the 5' end (Scheme S1). A 5'-phosphate is required for loading into the mid-domain of the Argonaut2 protein; however, natural phosphorylation of the siRNA by Clp1 kinase can be impacted by chemical modifications to the siRNA. 5'-vinylphosphonates, metabolically stable 5'-phosphate mimics, which have been shown to improve tissue accumulation and efficacy of conjugated siRNAs in vivo,^{30,31} were included in this experiment to reduce the potential impact of new modifications on activity. The unlocked sugar with 2'-O-methoxyethyl is believed to provide flexibility to load into the mid-domain of Argonaut2 along with steric bulk to protect against nucleases.

We administered 1 mg/kg (siRNA) doses to mice at Day 0, then sacrificed four animals from each treatment at each time point: 24 h, and 2, 4, 6, and 8 weeks. We measured *Mstn*

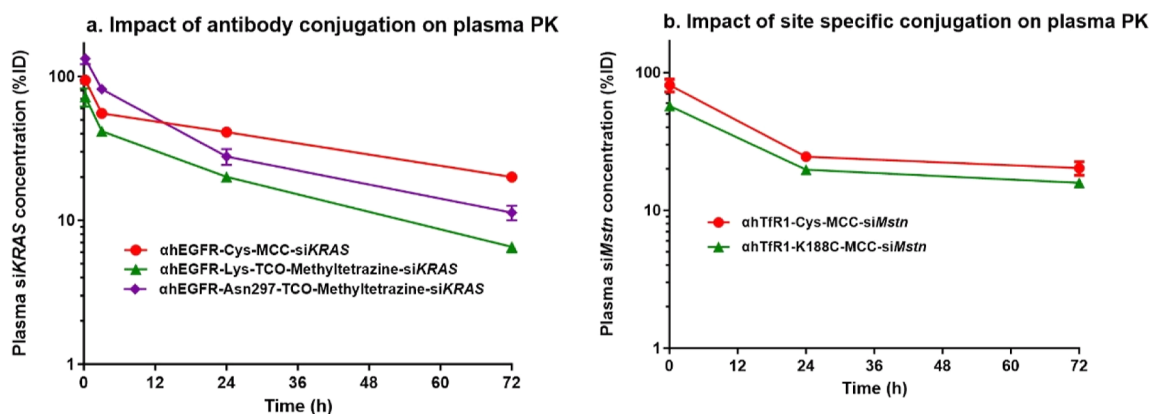


Figure 4. (a) Plasma PK analysis of α hEGFR-siKRAS AOCs conjugated to the mAb cysteine, lysine, and Asn297 as measured by percent of injected dose (% ID) versus time (h). (b) Mouse plasma PK of DAR1 α hTfR1-siMstn AOCs generated via random cysteine versus cysteine engineered at position 188 of the light chain.

mRNA expression and siRNA concentrations in the gastrocnemius muscle at each time point. Among the sugar modifications, UNA was found to be more stable than 2'-O-methyl, while LNA was more stable than UNA. This trend is true for siRNAs with no P=S on the termini of the sense strand (groups 1, 2, 4, and 6), as well as siRNAs that have one P=S at each end of the sense strand (groups 3, 5, and 7). The 5' end of the sense strand may be the most vulnerable to degradation in the absence of P=S groups, which can be seen by the reduced KD detected at Weeks 2 and 4. In the absence of P=S, the addition of 2 UNA or LNA (groups 4 and 6) at each termini of the sense strand improved KD at Weeks 2 and 4 compared with the 2'-O-methyl (group 2); however, by Weeks 6 and 8 it is clear that UNA and LNA in the absence of P=S are not sufficient due to the loss of KD. With all three sugar modifications (2'-O-methyl, UNA, and LNA) adding a single P=S to each side of the sense strand improves activity from Weeks 4 through 8, indicating that these more stable modifications can lead to longer durations of silencing. The two most active and stable modification patterns were found to be two P=S at each terminus of the sense and antisense strands as well as the pattern with two LNAs and one P=S at each terminus of the sense and two P=S at each terminus of the antisense strand; both maintain better than 80% KD over 8 weeks while all other patterns begin to show reduced KD by Week 8 (Figure 3). As demonstrated by Brown et al. (2020), stable siRNAs can remain in acidic compartments of the cell and act as a depot that results in continuous RISC loading beyond 2 weeks and sustained KD, despite decreasing siRNA concentrations in the tissue.³² Although the data are shown only for one antibody and siRNA conjugate, several other AOCs with different siRNA sequences and antibodies showed similar stability patterns in vivo. These findings highlight the importance of carefully designing and selecting chemical modification patterns to optimize siRNA stability and activity for therapeutic applications.

siRNA Conjugation to mAb Via Cysteine, Lysine, and Sugar Locations. Once we identified the optimal siRNA chemical modification pattern for in vivo delivery and long duration of action, we next evaluated the differences in various conjugation positions on the antibody, including cysteine, lysine, and sugar locations. We assessed differences in plasma PK in mice for antibody-siRNA conjugates comprised of an α hEGFR mAb, which does not cross-react with mouse EGFR,

conjugated to an siRNA targeting KRAS (siKRAS) at different sites on the antibody. The conjugation methods included random cysteine (TCEP reduction of disulfide followed by conjugation with siRNA-maleimide), random lysine (by nonselective addition of a TCO linker to lysine side chains), and glycan conjugation at Asn297 (sugar functionalization with TCO linker) (Figure 4a). AOCs were evaluated at a dose level of 0.5 mg/kg (siRNA). The full siRNA sequences for test articles evaluated in this experiment and throughout the manuscript are listed in Table S1.

Lysine and glycan conjugation of siRNA exhibited increased plasma clearance [46 and 11% lower area under the curve (AUC)], respectively, compared to cysteine conjugation (Figure 4a; SEM $n = 4$). The reduced plasma exposure of the lysine-conjugated siRNA is likely attributed to the greater heterogeneity and lower in vivo stability of lysine conjugates. While glycan conjugation is well tolerated for many antibody-drug conjugates (ADCs), it is not clear if the small reduction in plasma exposure of the glycan-conjugated AOC is the result of linker stability or the location of the siRNA on the antibody.

We also investigated the impact of random cysteine conjugation versus site-specific cysteine engineering on plasma clearance, since some locations of cysteine on the antibody show improved plasma PK of ADCs.³³ For random cysteine conjugation, we used α hTfR1-Cys-MCC-siMstn, while for site-specific cysteine engineering we used α hTfR1-K188C-MCC-siMstn, where the cysteine was engineered at position lysine 188 of the antibody light chain. No significant differences in plasma clearance were observed when conjugating siRNA to a random cysteine (Figure 4b).

While these data indicate cysteine conjugates may have advantages over glycan and lysine conjugates, additional studies are required to elucidate differences between PK, tissue delivery, and efficacy. Based on our data, additional results described herein will focus on random cysteine conjugation. Random cysteine conjugation was selected based on the relative ease of conjugate synthesis from most antibodies without the necessity of molecular engineering, less heterogeneity, clinical evaluation as shown by numerous ADCs, and manufacturability.

Drug to Antibody Ratio. We investigated the impact of DAR on AOC plasma clearance by assessing AOCs comprised of α hEGFR mAb conjugated to one, two, or three siRNAs targeting DMPK (α hEGFR-Cys-MCC-siDMPK). An antibody

that does not bind to a target in mouse was selected to show the impact of DAR on plasma clearance in the absence of target-mediated drug deposition. We observed clear trends of faster plasma clearance as a function of increasing DAR (Figure 5a). Figure 5b shows the clearance rate of DAR2 is nearly five

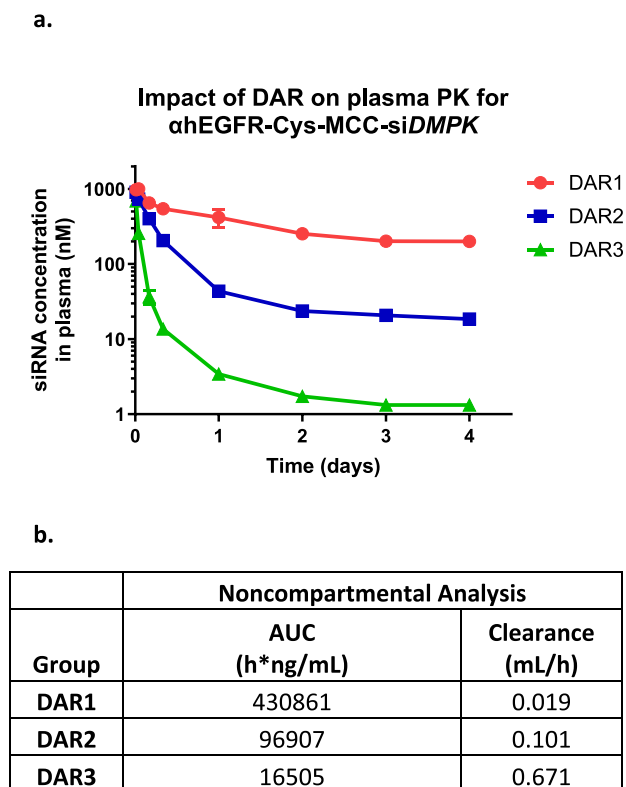


Figure 5. (a) Mouse plasma PK of AOCs (α hEGFR-Cys-MCC-siDMPK) comprised of α hEGFR mAb conjugated to one, two, or three siRNAs targeting *DMPK* mRNA. (b) Noncompartmental analysis of plasma PK data (0–96 h).

times greater than the clearance rate of DAR1 (0.101 vs 0.019 mL/h) calculated from 0 to 96 h. To determine if increasing

DAR impacts delivery to tissue or activity in muscle, α mTfR1-Cys-MCC-siMstn conjugates with a DAR of 1 and 2 were tested in mice. The rapid clearance from plasma observed with DAR2 is likely caused by greater nonproductive uptake by the liver. This rapid uptake by the liver leads to lower delivery and activity in the muscle (Figure S1). AOCs with a higher DAR may have altered physicochemical properties, such as increased size and significantly greater negative charge density, which can affect their interactions with plasma proteins and clearance mechanisms. Examples of the increased size and charge density of DAR2 conjugates include a left shift on SEC (Figure S2) and a right shift on strong anion exchange (Figure S9a). Protein interaction data are not available for these conjugates but binding of oligonucleotides has been observed by others.^{34,35}

Oligonucleotides themselves have been observed to interact strongly with plasma and cell surface proteins and it is possible that increasing the DAR may increase the avidity of these interactions.^{36–38} A similar phenomenon of rapid DAR2 clearance with ASO-antibody conjugates (ASOs are single strands and typically fully phosphorothioated oligonucleotides) has been observed by us (unpublished) and others.¹⁵ However, phosphorodiamidate morpholino oligonucleotide (PMO)-antibody conjugates (charge-neutral oligonucleotides) with up to DAR9.7 have no impact on plasma PK,³⁹ suggesting the charge and/or phosphorothioate content in siRNA and ASO DAR2 AOCs is driving clearance. Based on these data, we selected DAR1 for further evaluation throughout this paper.

Cysteine Conjugates with Cleavable Versus Non-cleavable Linkers. After selecting random cysteine conjugation as the conjugation strategy for antibody–siRNA conjugates, we focused on evaluating the impact of linker on plasma clearance. We evaluated AOCs using various cleavable and noncleavable linkers to assess their impact on in vivo stability as measured by plasma PK. To attain ideal pharmacokinetic profiles for active siRNA and desired effectiveness, it is necessary to regulate both the stability of antibody-conjugates and the release of siRNA. Favorable systemic stability is sought to safeguard linker-siRNA from hydrolysis while circulating. Simultaneously, efficient cleavage

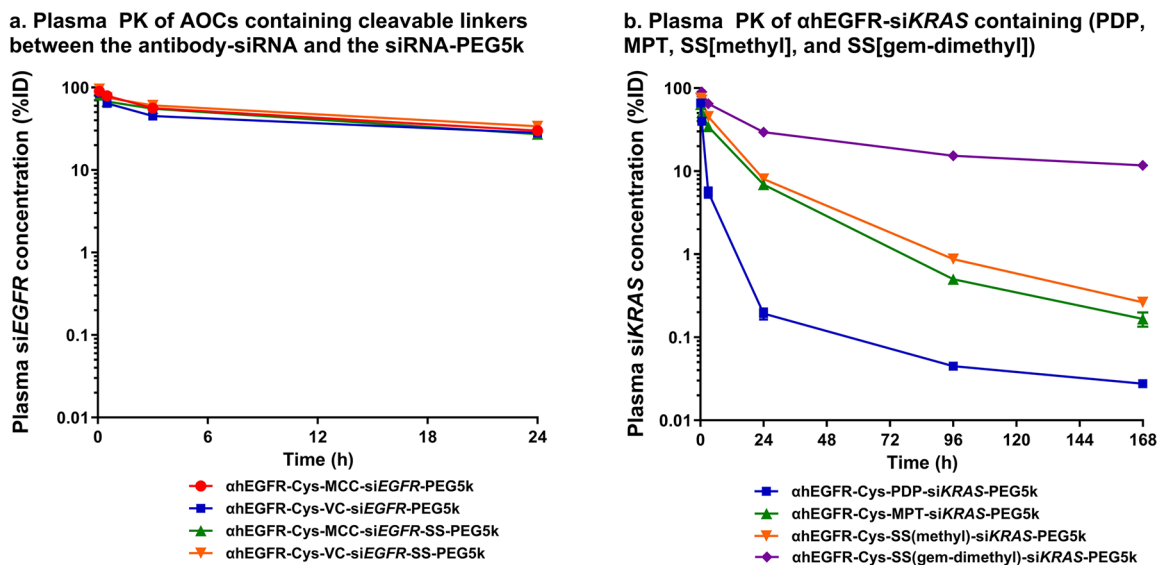


Figure 6. (a,b) Mouse PK of AOCs containing cleavable linkers.

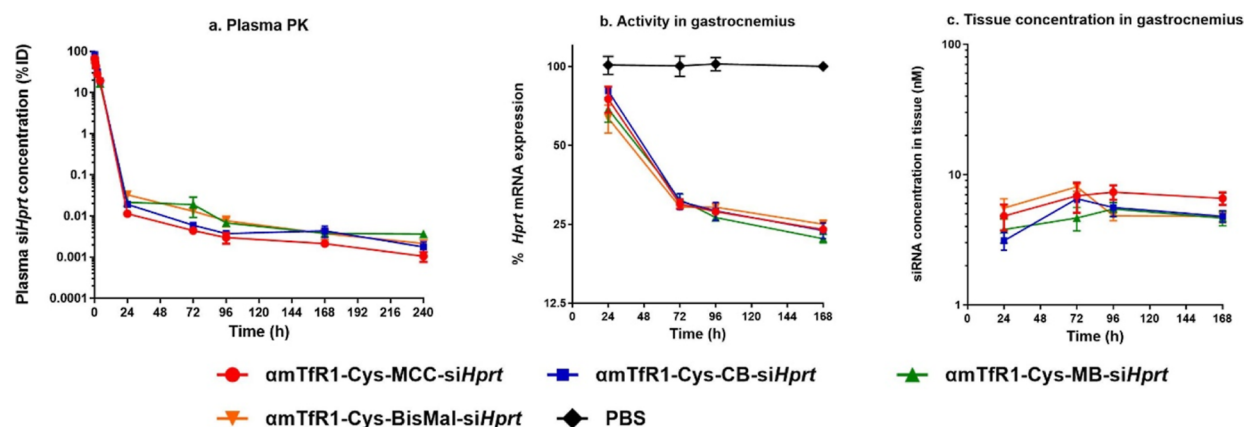


Figure 7. (a–c) PK/PD studies with α mTfR1-siHprt AOCs containing four different noncleavable linkers.

of linkers and release of siRNAs are crucial for accumulating a satisfactory quantity of active siRNA and meeting efficacy end points. Understanding these nuances is crucial for optimizing the design and functionality of antibody-siRNA conjugates. Table S2 summarizes the structures of the linkers. We assessed cleavable linkers including reducible disulfide (SS) linkers SS(methyl) and SS(gem-dimethyl) which exhibit increasing steric hindrance around the disulfide bond. Additionally, we evaluated a cathepsin-cleavable valine-citrulline (VC) linker, 3-(2-pyridyldithio)propionate (PDP) and methyl-(2-pyridyldithio)toluene (MPT). Noncleavable linkers included MCC, *m*-maleimidobenzoyl (MB), 4-((4-(cyanoethynyl)-benzoyl)oxy) (CB), and bismaleimide (BisMal).

The impact of these linkers on AOC plasma PK was tested on conjugates with a 5000 Da MW PEG (PEG5k) attached to the sense strand at the end opposite of antibody conjugation. The benefits of PEGylation for AOC stability are included in a subsequent section. We conducted mouse PK studies with AOCs comprised of an α hEGFR antibody conjugated to siEGFR-PEG5k (PEG5k conjugated to the 3' end of siEGFR by a stable thioether). We included a stable MCC linker between the mAb and siRNA (α hEGFR-Cys-MCC-siEGFR-PEG5k) and compared it with an enzyme-cleavable VC linker (α hEGFR-Cys-VC-siEGFR-PEG5k). Further iterations of these AOCs were synthesized to evaluate the impact of a cleavable SS linker between the siRNA and PEG5k (α hEGFR-Cys-siEGFR-SS-PEG5k and α hEGFR-VC-siEGFR-SS-PEG5k). We measured the plasma PK of these AOCs over time (Figure 6a,b) and observed no significant differences in clearance of the siRNA. However, our stem-loop assay for the detection of the antisense strand is insensitive to the presence of a PEG group attached to the sense strand, meaning the stability of the PEG5k group attached via a disulfide bond remains unclear. Furthermore, it is possible that a large PEG at the end of the siRNA could influence linker stability, for instance by limiting enzyme accessibility to the VC linker. However, considering the radius of gyration for PEG5k (~ 3 nm) and the length of a 21-nucleotide siRNA duplex (~ 7.5 nm), it is unlikely that PEG attached to the 3' end of the sense strand would have a significant impact on the stability of the linker at the 5' end. Further studies are needed to fully elucidate the impact of these modifications on the PK of AOCs.

Further mouse PK studies were conducted with AOCs comprised of α hEGFR mAb conjugated to siKRAS via different cleavable disulfide linkers (PDP, MPT, SS[methyl], and SS[gem-dimethyl]) between the mAb and siRNA.⁴⁰ These

AOCs each contained a PEG5k conjugated directly to the siKRAS. We dosed the AOCs at 0.5 mg/kg (siRNA) (Figure 6b). These results demonstrate that the linker stability matters for AOCs, and that increasing steric bulk around the disulfide (SS) linker is required to maintain plasma stability of these cleavable linkers. Incorporating the gem-dimethyl group into the disulfide linker reduced the clearance by more than 40 times (0.951 to 0.02 mL/h for PDP vs SS[gem-dimethyl] as shown in Table S3), suggesting that increased bulk slows down disulfide cleavage in circulation. Similar results have been observed with ADCs demonstrating the impact of disulfide linker length and steric hindrance on PK and efficacy.^{41,42} The rest of this manuscript will focus on noncleavable (stable) linkers other than the data shown in Figure 9.

Further Exploration of Non-cleavable Linkers (MCC, CB, MB, and BisMal). *α hEGFR-siKRAS AOC In Vitro Stability.* To compare the in vitro stability of different noncleavable linkers, we conjugated a human immunoglobulin (Ig) G1 antibody targeting EGFR (α hEGFR) to KRAS siRNA using four different noncleavable linkers (MCC, CB, MB, and BisMal). We tested the linker stability by observing siRNA release from the antibody after incubating it at 37 °C in PBS with 0.5 mM glutathione (Figure S3). Free antibody was measured by using strong-anion exchange high-performance liquid chromatography to indicate the loss of KRAS siRNA. Our results showed that α hEGFR-siKRAS AOCs containing CB, MB, and BisMal linkers were more stable than the MCC AOC in vitro, matching expectations from the literature.⁴³

Linker Impact on mTfR1 Antibody-Based siRNA Delivery to Muscles and Other Tissues. We synthesized and evaluated α mTfR1 antibody-siHprt conjugates via four different linkers (MCC, CB, MB, and BisMal) to understand the impact of in vitro linker stability differences on the AOC plasma PK, siRNA tissue delivery and target mRNA KD efficacy in a mouse PK and Hprt mRNA KD study. Despite differences in in vitro stability, we observed nearly identical plasma clearance profiles, as well as siRNA concentration and mRNA reduction in skeletal muscle, for all four conjugates (Figure 7). While others have shown that cytotoxic ADCs targeting tumors have improved activity when using stabilized linkers,⁴⁴ our data show that for α mTfR1 conjugates, there is no improvement in plasma exposure, tissue delivery, or KD when using linkers more stable than MCC. This result is best explained by Figure 7a showing greater than 80% of the dosed material is no longer in plasma after 4 h. This rapid target-mediated drug disposition is expected for a highly expressed receptor like TfR1 and

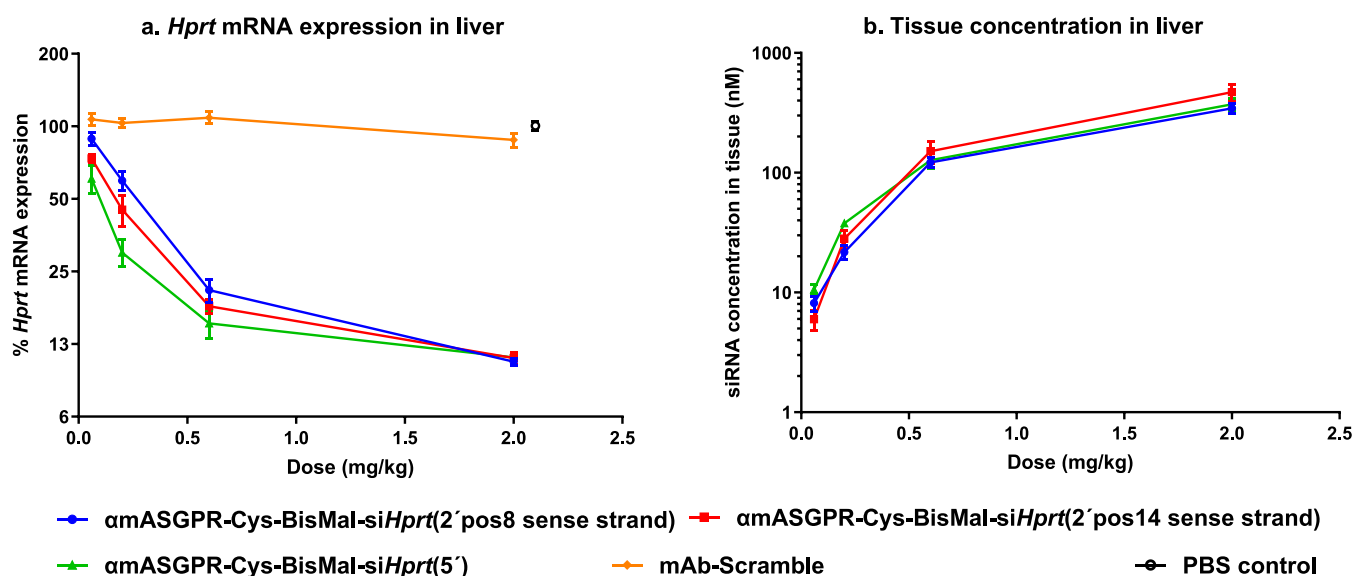


Figure 8. (a,b) Liver KD analysis of amASGPR-siHprt AOCs containing BisMal linkers to siRNA at the 5' end and positions (pos) 8 or 14 of the sense strand at 96 h postdose.

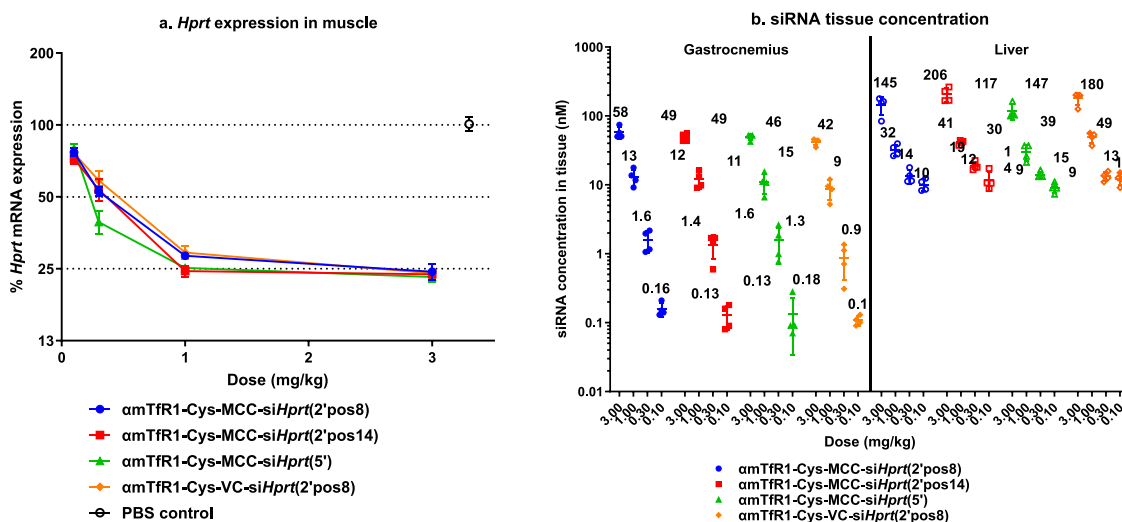


Figure 9. (a) Expression of *Hprt* in muscle and (b) siRNA tissue concentration after treatment with amTfR1-siHprt AOCs with MCC and VC linkers at various positions on the siHprt at 96 h postdose.

results in the majority of the AOC being taken into the tissue before any differences in linker stability could have an impact. It would be intriguing to explore whether these linker patterns persist with AOCs featuring extended plasma residence.

Linker Location on the siRNA. The spatial arrangement of the linker on the siRNA molecule could play a pivotal role in determining the stability of the siRNA itself. This influence is likely attributed to heightened steric hindrance, where the size and the proximity of the antibody creates obstacles to enzymes that can affect the stability of the siRNA as part of the conjugate as well as after release inside the cell. Understanding the impact of linker placement on siRNA stability is crucial for optimizing the design and performance of antibody-siRNA conjugates, especially in the context of therapeutic applications where stability is a critical factor for efficacy and duration of action. AOCs were synthesized to evaluate the effect of linker locations at the 3' or 5' end of the siRNA sense strand on the target mRNA KD in different tissue types. The AOCs we evaluated were comprised of an amTfR1 antibody conjugated

via an MCC linker to an siHprt at either the 3' or 5' end: amTfR1-Cys-MCC-N-5'-siHprt and amTfR1-Cys-MCC-N-3'-siHprt. We measured the KD of these AOCs in four tissues: muscle, heart, liver, and lung (Table S1 and Figure S4). These results demonstrate that the siRNA could be linked at either the 3' or 5' end of the sense strand without an impact on KD in the four tissues assessed. This also indicates that any remnant of the linker or the antibody at the 3' or 5' end of the siRNA sense strand does not impact RISC loading of duplex siRNA. These data show that the concentration of siRNA in tissue (Figure S4e) does not necessarily correlate with activity when comparing across tissues. While 20 nM concentrations in muscle generate approximately 70% KD, greater than 200 nM concentrations can be measured in the liver without achieving the same amount of activity. These data suggest that AOC activity can be impacted by siRNA delivery as well as cell type and, potentially, mechanism of siRNA internalization.

We synthesized additional AOCs to further evaluate the impact of linker positions in the middle of the siRNA sense

Table 1. Summary of Reaction Conditions and DAR Distributions for Various mAb Isotypes (%)

mAb isotype	TCEP eq	(linker) siRNA eq	DAR0 (%)	DAR1 (%)	DAR ≥ 2 (%)	isolated DAR 1 yield (%)
hIgG1	2	(MCC) 1	25	57	18	46
hIgG2	6	(MCC) 1	27	40	33	34
hIgG4	1.75	(MCC) 1.25	47	22	31	16
hIgG4	1.75	(BisMal) 1.25	46	31	23	25
Rat IgG2a	4	(MCC) 1.15	38	41	21	36
Rabbit IgG	4	(MCC) 1.1	26	41	33	35

strand. α hEGFR antibody was lysine conjugated to siKRAS with bicyclo[6.1.0]nonyne (BCN) linkers at siRNA positions 7, 13, or 17 on the sense strand off a 2' amine (Table S1 and Figure S5). Although we hypothesized that conjugating to the backbone of the siRNA would enhance protection against potential nucleases by antibody steric interference, we observed no impact on plasma PK based on the siRNA backbone linker location.

The impact of siRNA conjugation position on in vivo potency was tested in both liver and muscle with two different targeting antibodies. To assess the impact of siRNA conjugation positions on KD in the liver, we synthesized AOCs containing antimouse ASGPR mAb (α ASGPR) conjugated to siHprt via a BisMal linker at three different positions: 5' end versus positions 8 or 14 of the sense strand via 2' primary amine (Figure 8). We measured percent Hprt mRNA expression in the liver. The α ASGPR-Cys-BisMal-siHprt (5') demonstrated higher KD versus backbone positions 8 or 14 on the sense strand at lower doses, indicating potential interference in siRNA RISC loading with a linker attached at 2' locations. It is important to note that while conjugates at 2' positions 8 and 14 were less active at lower doses, they were able to achieve the same maximum KD as the 5' conjugate when the dose was increased to 2 mg/kg.

Additional AOCs were synthesized to evaluate the impact of siRNA conjugation sites on KD in the gastrocnemius. These AOCs were comprised of an α mTfR1 conjugated to siHprt with either a stable MCC or cleavable VC linker (Figure 9a). The MCC linker was evaluated at the 5' end as well as positions 8 and 14 on the sense strand via a 2' primary amine. Better gastrocnemius KD was observed for the MCC linker at the 5' end versus the backbone positions tested for the α mTfR1-siHprt AOCs at 0.2 mg/kg ($p < 0.05$), once again indicating the potential impact on the RISC loading of the released siRNA with linkers at positions 8 and 14 on the sense strand. An enzyme-cleavable VC linker was then evaluated at position 8 of the sense strand via a 2' primary amine. The α mTfR1-Cys-VC-siHprt (2'pos8 sense strand) AOC was synthesized to test the hypothesis that the self-immolating enzyme-cleavable linker would have the greatest impact at the center of the sense strand because of the potential for the noncleavable linker to have more steric bulk and prevent RISC loading. We based this hypothesis on in vitro research with similar modifications.¹² However, the AOC with the cleavable VC linker at position 8 of the sense strand showed lower activity compared with a noncleavable MCC linker 5' end conjugation at lower doses, indicating that propyl amine at 2' of position 8 of the siRNA sense strand is unlikely to be an optimal 2' substitution for RISC loading and target gene KD in muscle or liver. The VC linker cleavage kinetics for antibody-siRNA conjugates are not known. It is conceivable that the VC linker at position 8 could be more hindered than at the 5' end resulting in slower release of siRNA. The Hprt mRNA KD

reaches saturation beyond a dose of 1 mg/kg, yet the concentration of siRNA in tissues continues to rise (Figure 9b), indicating that receptor capacity is not the limiting factor. Our hypothesis to explain the absence of a dose response beyond 1 mg/kg, despite higher siRNA concentrations in tissues, is that Hprt is expressed in both myotubes and other cell types (such as fibroblasts and endothelial cells), whereas siRNA delivery primarily targets muscle cells.

We recognize the extensive work done around internucleotide triester, 2'-linker, and nucleobase conjugation approaches that identified SAR trends for GalNAc and other ligand conjugates.⁴⁵ Our study did not delve into these aspects; therefore, we caution against overgeneralization of our findings and recommend further studies to comprehensively understand the impact of siRNA conjugation sites on KD in different tissues and under varying conditions.

Impact of PEG on AOC Activity. PEG increases solubility, reduces nonspecific protein binding, provides protection from enzymes, and reduces clearance of various molecules such as proteins.⁴⁶ To assess the impact of PEG on the plasma PK of an AOC, we evaluated a set of AOCs with PEG on the inside versus the outside of siRNA (i.e., mAb-PEG5k-siRNA vs mAb-siRNA-PEG5k). The delivery and KD of PEG AOCs in mice with lymph node carcinoma of the prostate (LNCaP) xenograft were also assessed. These mice were treated with AOCs comprised of antihuman prostate-specific membrane antigen antibody (α hPSMA) conjugated to EGFR siRNA (siEGFR). The PEG was placed either between the mAb and siRNA or on the outside of the siRNA: α hPSMA-Cys-MCC-siEGFR, α hPSMA-Cys-MCC-siEGFR-PEG5k, and α hPSMA-Cys-PEG5k-siEGFR (Table S1 and Figure S6). These AOCs were designed to test the hypothesis that placing PEG on the outside of the siRNA would protect it from nucleases and increase its stability. Observations showed no significant change in plasma PK based on the size or placement of the PEG. However, placing PEG on the outside led to more KD and higher tissue concentration in the tumor, indicating a potential increase in the intracellular stability of siRNA. Furthermore, the placement of PEG5k externally is likely to have an impact on the hydrophilicity of the conjugate. This could influence the tumor-to-liver ratio and subsequent mRNA reduction. In addition, PEG might shield some of the siRNA charge, which could further affect its interaction with cellular components and its overall effectiveness. These concepts are well described in ADCs.⁴⁶

To further evaluate the potential impact of PEGylation on siRNA in the muscle, AOCs were synthesized using an α mTfR1 antibody conjugated at the cysteine via an MCC linker to an siCtnnb1 sense strand with and without PEG5k on the outside (Figure S7). Tissue PK data of these α mTfR1-siCtnnb1 AOCs suggest that PEG5k attached to the 3' end of siCtnnb1 sense strand (while conjugating the 5' end to the antibody) increases the amount of siRNA detected in

gastrocnemius as well as *Ctnnb1* mRNA KD. Specifically, a 1 mg/kg dose with PEG achieved the same tissue concentration as the 3 mg/kg dose without PEG. Moreover, the KD at 1 mg/kg with PEG was comparable to the KD at 3 mg/kg without PEG.

Our data indicate that including PEG in the AOC architecture could help with siRNA tissue PK, likely resulting in increased gene KD efficacy. Further research is needed to understand the optimal size of the PEG needed in these AOCs.

Applicability of siRNA Bioconjugation across Multiple Antibody Isotypes. We can apply siRNA bioconjugation to a variety of antibody isotypes, including human (h) IgG1, hIgG2, hIgG4, rat IgG2a, and rabbit IgG. The structure of the antibody and conditions used for partial reduction determine the conjugation efficiency and DAR distribution of random cysteine conjugation of siRNA. Each antibody isotype requires different ratios of reagents to maximize DAR1, and IgG4 predominantly gives DAR2 (Table 1). We evaluated antibodies for conjugation, including antimouse TfR1, anti-PSMA, anti-EGFR, and antimouse ASGPR. Interestingly, for hIgG4, the use of a monofunctional versus bifunctional linker (MCC, BisMal) influenced the ratio of DAR1 and $\text{DAR} \geq 2$. This linker-dependent change in DAR distribution may be due to the Fab arm exchange observed in IgG4 antibodies and the ability of the BisMal linker to stabilize the DAR1 structure. While DAR1 yields may vary greatly across antibody isotypes, we can obtain sufficient yields of pure DAR1 for preclinical research. The observed variation in DAR distribution between antibody isotypes for AOCs is consistent with the differences observed with ADCs generated with different isotypes.⁴⁷ This correlation is anticipated, given the similar conditions used for partial reduction of the antibodies. However, AOCs present a unique challenge as they require the isolation of only DAR1. This is further complicated by the high MW and negative charge of siRNAs which creates steric and charge repulsion that can slow reactions for less exposed positions on the antibody.

CONCLUSIONS

AOCs are complex bioconjugates and the data herein illustrate a comprehensive SAR evaluation of siRNA AOCs. This evaluation has allowed us to optimize siRNA chemistries, select linkers, identify points of conjugation on the antibody and siRNA, develop strategies to conjugate an antibody to an siRNA oligonucleotide, and understand the impact of oligonucleotide-to-antibody ratios on AOC activity. Our findings show the importance of selecting appropriate siRNA chemical modifications to improve tissue delivery in vivo and achieve higher target gene KD in tissues of interest. Our platform can accommodate various points of conjugation on the antibody, offering flexibility to choose between cysteine, site-specific cysteine, and sugar conjugations. Initial studies indicate that lysine conjugates are not optimal for AOCs, but additional studies are needed to understand the differences. The siRNA-to-antibody ratio is limited to 1, likely due to high charge of each siRNA and the potential phosphorothioate-mediated nonspecific binding of $\text{DAR} \geq 2$ AOCs. The AOC platform can accommodate most noncleavable linker chemistries when conjugated on both ends of the sense strand, indicating that remnants of linker attached to the released siRNA sense strand, after AOC digestion in lysosomes, may not interfere with Ago2 protein binding and RISC loading. We found no significant differences in plasma PK, siRNA tissue

delivery, and target gene KD between noncleavable linkers like MCC, BisMal, CB, and MB with rapidly internalized αmTfR1 AOCs. Additional evaluation with AOCs that show longer plasma PK are needed to tease out any differences between these noncleavable linkers. Results from our evaluation of cleavable linkers demonstrate that AOCs with less sterically hindered disulfide linkers like PDP, MPT and SS(methyl) are less stable than more sterically hindered disulfide SS(gem-dimethyl). The reduced plasma clearance of AOCs correlated with the increasing steric hindrance around the disulfide bond. The VC linker offered no significant improvement over stable MCC or BisMal linkers, indicating the tolerance for remnant linker attached to siRNA for Ago2 binding and RISC loading. The exploration of conjugation positions, other than 5' or 3' end positions on the sense strand, on both strands of siRNA also revealed that the internal conjugation locations show lower activity than the 5' end conjugation. This is likely due to interference of siRNA duplex interaction with either Ago2 protein or RISC loading. Addition of 5000 molecular weight PEG on the other side of siRNA in an AOC, by sandwiching siRNA between antibody and PEG, showed approximately three-fold improved tissue delivery to the targeted tumor cells with αhPSMA AOC and a five-fold increase in muscle with αmTfR1 AOC. The increased delivery of siRNA resulted in higher target gene KD. PEG is likely providing steric shielding to nucleases and may reduce nonspecific protein binding of AOC.

Overall, our SAR study provides important insights into the design and optimization of siRNA AOCs, identifying key factors that can influence PK, tissue delivery, and efficacy of these complex bioconjugates. These findings are consistent across multiple antibody targets and siRNAs, suggesting these design principles can be applied to the entire AOC class. By understanding the SAR of AOCs, researchers can continue to develop new and improved therapeutics for a range of diseases and conditions.

EXPERIMENTAL SECTION

siRNA Synthesis. The oligonucleotide siRNA sequences were assembled on solid phase using well-described and standard phosphoramidite methodology. Single strands were synthesized on a MerMade 12 synthesizer (BioAutomation) with phosphoramidite monomers (2'OMe, 2'F, LNAs, inverted abasic, C_6NH_2 , and $\text{C}_6\text{SSC}_6\text{dT}$) and universal controlled pore glass solid support purchased from commercial sources. The vinylphosphonate UNA-MOE-U phosphoramidite monomer was synthesized in-house as per the synthesis shown in Scheme S1.⁴⁸ All phosphoramidites were used at a concentration of 100 mM in 100% anhydrous acetonitrile utilizing a 2.5 molar excess of ETT activator in acetonitrile with 2×6 min coupling time. Oxidation of phosphite linkages was achieved using 0.02 M I_2 in 9:1 acetonitrile/ H_2O for 60 s. Sulfurization was achieved with 0.05 M DDTT (3-dimethylaminomethylene amino-3H-1,2,4-dithiazole-5-thione) in 60/40 pyridine/acetonitrile for 6 min per cycle. Capping was performed with tetrahydrofuran (THF)/pyridine/acetic anhydride for 60 s and 16% methyl imidazole in THF for 60 s per cycle. Deblocking was performed with 3% trichloroacetic acid in acetonitrile for 2×40 s per cycle. Cleavage of siRNA from solid support and deprotection of all protecting groups was accomplished using a 1:1 ratio of 28% aqueous ammonia and 40% methylamine in methanol for 2 h at room temperature. Crude oligonucleotides were purified using strong anion exchange with phosphate buffers (pH 7.2) containing sodium bromide; the desired fractions were chosen based on LCMS analysis, then pooled and desalted. The identification and purity of the single strands were determined by LCMS and RP-HPLC followed by lyophilization. The single strands were then dissolved in

H₂O and equal molar amounts of antisense and sense strand were annealed by heating to 65 °C for 5 min and cooling to room temperature to form the final duplex, which was analyzed by anion exchange chromatography.

Characterization of Oligonucleotides. Single-strand purities were measured by ion-pair RP-HPLC using an XBridge C18 OST column (Waters, Milford MA; 2.1 × 50 mm, 2.5 μm) with a gradient from 6 to 35% mobile phase B over 15 min at a flow rate of 0.25 mL/min and column temperature of 65 °C. Mobile phase A consisted of 100 mM 1,1,1,3,3,3-hexafluoroisopropanol (HFIP), 16.3 mM triethylamine (TEA), and 1% methanol. Mobile phase B consisted of 100 mM HFIP, 16.3 mM TEA, and 95% methanol. All single strands were at least 85% pure. Single-strand identities were confirmed by mass spectrometry (MS), on LCQ Deca XP + Ion Trap MS (Thermo Finnigan, San Jose, CA), used in line with ion-pair reversed-phase HPLC. The MS was run in negative ion mode with a capillary temperature of 315 °C and voltage of −100 V. Data analysis was performed with ProMass deconvolution software (Novatia, Newtown, PA). Duplex purity was measured by anion exchange chromatography using a DNAPac-200 column (Thermo Fisher, Waltham, MA; 4 × 250 mm) with a gradient of 15–58% mobile phase B over 9 min with a flow rate of 1 mL/min. Mobile phase A contained 20 mM sodium phosphate buffer and 20% ethanol; mobile phase B consisted of mobile phase A plus 800 mM NaClO₄. All duplexes were at least 90% pure.

siRNA-PEG Synthesis. The siRNA sense strands used for conjugation were synthesized with a C6 amino or C6 sulfur linker at the 5′- or 3′-end, or 2′-internal modifications (C3 amino). siRNAs with 3′ sense strand C₆SSC₆DT modifications were dissolved in 100 mM phosphate buffer pH 7.4 then reduced with 20 eq TCEP at room temperature. Excess TCEP was removed using ultrafiltration into 20 mM phosphate buffer with 150 mM sodium chloride then added to 3 equiv of maleimide-PEG5k or 5 equiv of orthopyridyl disulfide PEG5k (Laysan Bio, Arab AL) for a reducible linkage. Resulting siRNA-PEG was purified using the strong anion exchange chromatography method described for AOCs below. The desired fractions were then isolated using ultrafiltration with 3 kDa molecular weight cutoff (MWCO). siRNA-PEG purity was measured by anion exchange chromatography using an Agilent 1200 with a Thermo Fisher ProPac SAX-10 BioLC column (250 × 4 mm) at ambient temperature. siRNA-PEG was separated from unconjugated siRNA and excess PEG with a gradient of 10–60% mobile phase B over 8 min and flow rate at 0.75 mL/min, monitoring at 260 nm. Mobile phase A was 10 mM Tris Base pH 7.2 with 20% ethanol, mobile phase B contained A plus 1.5 M sodium chloride. All siRNA-PEG purities were greater than 95%.

Linker-siRNA Generation. siRNA duplex in 50 mM phosphate buffer at a concentration of 100 mg/mL was pH adjusted with sodium hydroxide to pH 7.4. The siRNA was then diluted with dimethyl sulfoxide (DMSO) to achieve a final DMSO concentration of 50% once the linker was added. Linker (10 equiv to the siRNA) dissolved in DMSO was added to the siRNA and mixed at room temperature for 30 min. The progress of the reaction was monitored by MS, and the absence of a sense strand indicated the completion of the conjugation. Excess linker was removed using ultrafiltration with a 3 kDa MWCO. Linker-siRNA purity was not determined and was used in situ. Disulfide linkers used were succinimidyl 3-(2-pyridyldithio)propionate (SPDP), 4-succinimidylloxycarbonyl- α -methyl- α -(2-pyridyldithio)toluene (SMPT), SS[methyl] (2,5-dioxopyrrolidin-1-yl 4-(pyridine-2-yl)disulfanyl)pentanoate, and SS[gem-dimethyl] 2,5-dioxopyrrolidin-1-yl 4-methyl-4-((5-nitropyridin-2-yl)disulfanyl)pentanoate (Annova Chem, San Diego, CA). Maleimide linkers used were NHS-PEG5k-maleimide (Laysan Bio, Arab, AL), maleimidocaproyl-L-valine-L-citrulline-*p*-aminobenzyl alcohol *p*-nitrophenyl carbonate (MC-Val-Cit-PAB-PNP) (Broadpharm, San Diego, CA), succinimidyl-4-(*N*-maleimidomethyl)cyclohexane-1-carboxylate (SMCC) (Thermo Fisher, Waltham, MA), bis-maleimide-dPEG4-TFP ester (BisMal) (Quanta Bioscience, Plain City, OH), sodium 4-((4-(cyanoethynyl)benzoyl)oxy)-2,3,5,6-tetrafluorobenzenesulfonate (CBTF), and MB-*N*-hydroxysuccinimide ester (Sigma, St. Louis, MO). Click linkers used were TCO-PEG4-NHS, NHS-PEG1-Azide

(Broadpharm, San Diego, CA), and methyltetrazine-PEG4-NHS (Sigma, St. Louis, MO).

Cysteine, Lysine, and Glycan Conjugations. Cysteine-conjugated AOCs were generated using a standard random cysteine conjugation method. The interchain disulfide bonds of the antibody were partially reduced with TCEP (Thermo Fisher, Waltham, MA) at 37 °C for 2–4 h prior to conjugation with a maleimide linker-oligonucleotide (molar equivalents of TCEP and siRNA-linker required for conjugation to each antibody isotype are listed in Table 1). Lysine-conjugated AOCs were generated using a standard random lysine conjugation method (with the degree of conjugation relative to the antibody). The antibody at 10 mg/mL in PBS (pH 7.4) was incubated with 2 equiv relative to antibody of NHS-linker (TCO, maleimide, azide) at room temperature for 90 min. Excess linker was removed using 50 kDa MWCO ultrafiltration prior to conjugation with 1.75 equiv of a functionalized (tetrazine, sulfhydryl, BCN)-oligonucleotide. Glycan-conjugated AOCs were generated using Asn297 TCO modified antibody from Fisher Scientific, using SiteClick and conjugated with a methyl tetrazine-oligonucleotide. All reaction mixtures after the conjugation were purified on fast protein liquid chromatography using a preparative anion exchange purification method to isolate the DAR1 antibody-siRNA conjugate.

Preparative Anion Exchange Purification. SAX purification was performed on an ÄKTA Pure system with a Sartobind Q (Sartorius, Göttingen, Germany; Nano 3 mL with 8 mm bed height) at ambient temperature with a gradient of 20–40% mobile phase B over 20 column volumes and a flow rate of 3 mL/min, monitoring at ultraviolet (UV) 220, 260, and 280 nm. Mobile phase A was 20 mM sodium phosphate buffer pH 7.2; mobile phase B contained A plus 1.5 M sodium chloride (Figure S8).

Overview of Cysteine Conjugation Using NEM Capping with or without DHAA Oxidation. AOCs (*α*hEGFR-siKRAS) were generated via cysteine conjugation. Cysteine conjugates were made by partially reducing antibody interchain disulfide bonds with TCEP to react with maleimide on the siRNA. On human IgG1 antibodies, the primary location of siRNA conjugation is on the cysteine of the heavy chain previously forming the heavy chain-light chain disulfide.⁴⁷ During this process, some cysteine remained unreacted.

Overview of Free Cysteine Determination. The average number of free cysteines on DAR1 AOCs was estimated by quantifying the amount of fluorescein-5-maleimide (AnaSpec, Fremont, CA) able to react with the antibody relative to a standard curve measured by SEC-HPLC. For each replicate, 0.25 mg of mAb or AOC in PBS at 5 mg/mL was incubated with 5 mol equiv of fluorescein-5-maleimide (5 μL at 0.715 mg/mL in DMSO) at room temperature for 1 h protected from light (5 equiv was selected based on optimization experiments to determine the amount of fluorescein-5-maleimide needed to saturate the free cysteine reaction on mAb reduced with 1.5 EQ of TCEP). After a 3× dilution, 30 μL of each sample was tested by SEC-HPLC (Yarra SEC-3000 column, 1 mL/min flow rate, mobile phase: phosphate buffer +300 mM NaCl) the absorbance at 495 nm was measured and the AUCs for peaks corresponding to the antibody or DAR1 AOC were calculated and compared with a standard curve of fluorescein-5-maleimide to quantify the number of cysteine available for reaction.

Purity Statement. AOCs with a DAR of 1 are at least 95% pure by strong anion exchange chromatography and SEC. The single-strand oligonucleotides used to make the duplex siRNAs for conjugation were at least 85% pure by the appropriate HPLC methods and identified by MS.

Animal Studies. All animal procedures were approved by the Institutional Animal Care and Use Committee of Explora BioLabs in accordance with guidelines of the Association for Assessment and Accreditation of Laboratory Animal Care. Mice were housed in ventilated cages under a 12 h light-dark cycle and fed ad libitum with standard rodent chow. All in vivo experiments were conducted using four mice per group, each weighing ~24 g.

In vivo studies in wild-type mice were performed in female Hsd/ICR (Envigo) mice. All the dosing was based on the siRNA weight, not the total conjugate (1 mg/kg = 1 mg siRNA/kg). Mice were

dosed intravenously at 5 mL/kg. At end of study, mice were euthanized and tissues removed for analysis. For the assessment of delivery to tumors, male Crl:SHO-PrkdcscidHrhr (Charles River) mice were inoculated in the right flank subcutaneously with 7 million LNCaP cells in a mixture of 0.2 mL total of RPMI 1640 with matrigel (1:1). Mice were randomly grouped when tumors reached 200 mm³ and given an intravenous dose of the test article at 5 mL/kg. At end of study, or when tumors reached 2000 mm³, mice were euthanized and tissues removed for analysis.

Pharmacodynamic Evaluation. Tissue was processed for RNA extraction using the Zymo Direct-zol-96 RNA purification kit (Zymo Research Corporation, Irvine, CA). Once RNA quality was confirmed, cDNA was generated from purified RNA and used in quantitative reverse transcription polymerase chain reaction analysis for gene expression for each of the targets of interests.⁴⁹ The reference mRNA transcripts for Ppib were used in $\Delta\Delta C_t$ calculations for tissue samples.⁵⁰

To evaluate siRNA concentrations in plasma or tissues, custom stem-loop-RT-qPCR assays were developed as described previously.⁵¹ For each siRNA, a specific stem-loop-RT-qPCR assay was designed to quantify the antisense strand, using custom DNA forward, reverse, and reverse transcription primers (Integrated DNA Technologies, Coralville, IA) and a custom Taqman probe (Thermo Fisher, Waltham, MA). Noncompartmental analysis was performed on individual mouse plasma total siRNA concentration versus time data to estimate PK parameters for AUC and clearance using Phoenix WinNonlin v8.3 (Certara, Princeton, NJ). An intravenous bolus administration model (Plasma 200–202) was used and AUC was calculated using the “linear-up log-down” calculation method.

AOC Characterization. The isolated DAR1 antibody–siRNA conjugate was then characterized by SAX, SEC, reduced capillary gel electrophoresis (RCGE), UV, and MS.

Analytical Anion Exchange Chromatography. After bioconjugation, the DAR distribution of the AOC reaction mixture was analyzed by SAX. SAX analysis was performed on an Agilent 1200 with Thermo Fisher ProPac SAX-10 BioLC column (250 × 4 mm) at ambient temperature with a gradient of 10–60% mobile phase B over 8 min and flow rate at 0.75 mL/min, monitoring at UV 220, 260, and 280 nm. Mobile phase A was 10 mM Tris Base pH 7.2 with 20% ethanol; mobile phase B contained A plus 1.5 M sodium chloride. DAR1 was then isolated by column chromatography (Figure S9a). The DAR1 final product was then characterized by SAX [same method as described above; UV (Figure S9b)], SEC (Figure S9c), RCGE (Figure S9d), and MS (Figure S9e).

UV Spectroscopy. AOC concentrations were determined using the absorbance at 280 nm and an extinction coefficient of 2.28 L·g^{−1}·cm^{−1} for DAR1 and 3.43 L·g^{−1}·cm^{−1} for DAR2. Absorbance ratios at 260 nm/280 nm were also measured to confirm DAR. The extinction coefficient used for DAR1 and DAR2 AOCs was calculated using the UV absorbance at 280 nm and the BCA measurement of antibody concentration (compared to a standard curve with the same unconjugated antibody) of more than 20 different purified conjugates. The values were within 13 and 9% (DAR1 and DAR2) of the values calculated using the mAb and siRNA extinction coefficients and the formula below. The concentration of the mAb portion of an AOC was used to calculate siRNA concentration and total AOC concentration using the siRNA MW of 14,100 Da and antibody MW of 150,000 Da. A 1 mg/kg siRNA dose corresponds to 10.6 mg/kg mAb dose and 11.6 mg/kg AOC dose.

$$[\text{mAb}] = \frac{A_{280, \text{mAb}}}{\epsilon_{280, \text{mAb}} + \text{DAR} \epsilon_{280, \text{siRNA}}}$$

siRNA single strand concentrations were determined by using the absorbance of each strand at 260 nm and an extinction coefficient estimated using the nearest neighbor method. Duplex siRNA concentrations were calculated based on the known amount of each single strand used and the appropriate dilutions. When needed, duplex extinction coefficients were estimated using the formula below

including a correction factor of 0.8 to account for the hypochromic effect of duplexed nucleic acids.

$$\epsilon_{260, \text{siRNA}} = (\epsilon_{260, \text{antisense strand}} + \epsilon_{260, \text{sense strand}}) \times 0.8$$

Size Exclusion. SEC analysis was performed on an Agilent 1200 with a Yarra 3 μm SEC-300 column (300 × 7.8 mm) at ambient temperature with an isocratic flow of 55 mM potassium phosphate monobasic 62 mM sodium phosphate dibasic 100 mM sodium sulfate with 0.05% sodium azide pH 6.7 buffer with a flow rate of 1 mL/min for 15 min, monitoring at UV 260 and 280 nm.

Reduced Capillary Electrophoresis. Reduced capillary electrophoresis-sodium dodecyl sulfate was performed on ProteinSimple Maurice Plus instrument using Maurice CE-SDS PLUS cartridges according to manual instructions. Samples were prepared to 1 mg/mL by protein concentration. Internal standard and 2-mercaptoethanol were added, then heated to 70 °C for 10 min. Reduced samples were run with a sample load of 20 s at 4600 V, and separation for 25 min at 5750 V.

MS. Intact MS was performed by Thermo Fisher with a Q Exactive BioPharma. 5 μg of sample was loaded onto an Acquity UPLC protein BEH SEC column (4.5 × 100 mm, 1.7 μm) and eluted with 50 mM ammonium acetate at a flow rate of 0.3 mL/min and a column temperature of 55 °C. MS was run with a spray voltage of 4.2 kV and an ion transfer tube temperature of 320 °C.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jmedchem.4c00802>.

Please see siRNA AOC Supplement. siRNA sequence and chemical modifications for evaluated test articles; siRNA conjugation comprehensive guide; noncompartmental analysis of plasma PK data in Figure S5b; plasma PK, tissue concentration, and mRNA expression of $\alpha\text{mTfR1-Cys-MCC-siMstn}$ conjugates; DAR1 versus DAR2 SEC comparison; in vitro stability of $\alpha\text{hEGFR-siKRAS}$ AOCs; KD and tissue concentration of $\alpha\text{mTfR1-siHprt}$ AOCs; plasma PK of $\alpha\text{hEGFR-siKRAS}$ AOCs; tumor delivery and KD of $\alpha\text{hPSMA-siEGFR}$ AOCs; PK of $\alpha\text{mTfR1-siCtnnb1}$ AOCs; purification of antibody–siRNA conjugation reaction mixture by SAX; purification and characterization of DAR1 AOC; UNA vinylphosphonate (vpUq) amidite synthesis (PDF)

■ AUTHOR INFORMATION

Corresponding Author

Venkata Ramana Doppalapudi – Avidity Biosciences, Inc., San Diego, California 92121, United States;
Email: ramana@aviditybio.com

Authors

Michael Cochran – Avidity Biosciences, Inc., San Diego, California 92121, United States
Danny Arias – Avidity Biosciences, Inc., San Diego, California 92121, United States
Rob Burke – Avidity Biosciences, Inc., San Diego, California 92121, United States; Present Address: Seawolf Therapeutics, One Sansome Street Suite 3630, San Francisco, CA 94104, USA
David Chu – Avidity Biosciences, Inc., San Diego, California 92121, United States; Present Address: Capstan Therapeutics, 9880 Campus Point Drive, San Diego, CA 92121.
Gulin Erdogan – Avidity Biosciences, Inc., San Diego, California 92121, United States

Michael Hood – Avidity Biosciences, Inc., San Diego, California 92121, United States
Philip Kovach – Avidity Biosciences, Inc., San Diego, California 92121, United States
Hae Won Kwon – Avidity Biosciences, Inc., San Diego, California 92121, United States
Yanling Chen – Avidity Biosciences, Inc., San Diego, California 92121, United States
Michael Moon – Avidity Biosciences, Inc., San Diego, California 92121, United States; Present Address: Amgen Inc., 750 Gateway Blvd., Suite 100, South San Francisco, California 94080.
Christopher D. Miller – Avidity Biosciences, Inc., San Diego, California 92121, United States; Present Address: California North state University College of Medicine, 9700 W Taron Dr, Elk Grove, CA 95757.
Hanhua Huang – Avidity Biosciences, Inc., San Diego, California 92121, United States
Arthur Levin – Avidity Biosciences, Inc., San Diego, California 92121, United States

Complete contact information is available at:
<https://pubs.acs.org/10.1021/acs.jmedchem.4c00802>

Author Contributions

The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

Notes

The authors declare the following competing financial interest(s): At the time this research was conducted, all authors were employees of Avidity Biosciences and may have equity in the company.

ACKNOWLEDGMENTS

Authors acknowledge John Manhard for generating antibodies with site-specific cysteine conjugation handles. Thanks to Baorui Lin and Zhanlei Wei of WuXi TIDES for their work on the synthesis and scale-up of vp-UNA-3'-phosphoramidite. Special thanks to Son Lam and Han Cho for technical review of the manuscript. The authors would like to thank Gemma Hall, DPhil, and Megan Hotard Jarrell, MS, from Lighthouse Medical Communications, New York, NY, USA, for providing medical writing support. Medical writing support was funded by Avidity Biosciences, San Diego, CA, USA, in accordance with Good Publication Practice (GPP3) guidelines.

ABBREVIATIONS

% ID, percent of injected dose; ADC, antibody–drug conjugate; Ago2, Argonaute2; AOC, antibody–oligonucleotide conjugate; ASGPR, asialoglycoprotein receptor; ASO, antisense oligonucleotide; BCN, bicyclo[6.1.0]nonyne; BisMal, bismaleimide; CBTF, sodium 4-((4-(cyanoethynyl)benzoyl)-oxy)-2,3,5,6-tetrafluorobenzenesulfonate; DAR, drug-to-antibody ratio; DHAA, dehydroascorbic acid; EGFR, epidermal growth factor receptor; Eq, equivalent; GalNAc, N-acetylgalactosamine; HFIP, 1,1,1,3,3,3-hexafluoroisopropanol; HPLC, high-performance liquid chromatography; KD, knockdown; LNA, locked nucleic acid; LNCaP, lymph node carcinoma of the prostate; mAb, monoclonal antibody; MBS, *m*-maleimido-benzoyl-*N*-hydroxysuccinimide ester; MS, mass spectrometry; NEM, *N*-ethylmaleimide; NHS, *N*-hydroxysuccinimide; P=S, phosphorothioates; PBS, phosphate buffered saline; PEG,

polyethylene glycol; PSMA, prostate-specific membrane antigen; RCGE, reduced capillary gel electrophoresis; RISC, RNA-induced silencing complex; RT-qPCR, quantitative reverse transcription polymerase chain reaction; SAR, structure–activity relationship; SAX, strong anion exchange chromatography; SEC, size exclusion chromatography; siRNA, small interfering RNA; SMCC, succinimidyl-4-(*N*-maleimidomethyl)cyclohexane-1-carboxylate; SMPT, 4-succinimidylloxycarbonyl- α -methyl- α -(2-pyridyldithio)toluene; SPDP, succinimidyl 3-(2-pyridyldithio)propionate; SS, disulfide; TCEP, tris (2-carboxyethyl) phosphine; TEA, triethylamine; TfR1, transferrin receptor 1; UNA, unlocked nucleic acid; VC, valine-citrulline

REFERENCES

- (1) Arslan, F. B.; Ozturk, K.; Calis, S. Antibody-Mediated Drug Delivery. *Int. J. Pharm.* **2021**, 596, 120268.
- (2) Awwad, S.; Angkawinitwong, U. Overview of Antibody Drug Delivery. *Pharmaceutics* **2018**, 10 (3), 83.
- (3) Dovgan, I.; Koniev, O.; Kolodych, S.; Wagner, A. Antibody-Oligonucleotide Conjugates as Therapeutic, Imaging, and Detection Agents. *Bioconjugate Chem.* **2019**, 30 (10), 2483–2501.
- (4) Ovacik, M.; Lin, K. Tutorial on Monoclonal Antibody Pharmacokinetics and Its Considerations in Early Development. *Clin. Transl. Sci.* **2018**, 11 (6), 540–552.
- (5) Chery, J. RNA Therapeutics: RNAi and Antisense Mechanisms and Clinical Applications. *Postdoc J.* **2016**, 4 (7), 35–50.
- (6) Levin, A. A. Treating Disease at the RNA Level with Oligonucleotides. *N. Engl. J. Med.* **2019**, 380 (1), 57–70.
- (7) Burnett, J. C.; Rossi, J. J. RNA-Based Therapeutics: Current Progress and Future Prospects. *Chem. Biol.* **2012**, 19 (1), 60–71.
- (8) Crooke, S. T.; Witztum, J. L.; Bennett, C. F.; Baker, B. F. RNA-Targeted Therapeutics. *Cell Metab.* **2018**, 27 (4), 714–739.
- (9) Hu, B.; Zhong, L.; Weng, Y.; Peng, L.; Huang, Y.; Zhao, Y.; Liang, X.-J. Therapeutic siRNA: State of the Art. *Signal Transduct. Target. Ther.* **2020**, 5 (1), 101.
- (10) Egli, M.; Manoharan, M. Chemistry, Structure and Function of Approved Oligonucleotide Therapeutics. *Nucleic Acids Res.* **2023**, 51 (6), 2529–2573.
- (11) Janas, M. M.; Schlegel, M. K.; Harbison, C. E.; Yilmaz, V. O.; Jiang, Y.; Parmar, R.; Zlatev, I.; Castoreno, A.; Xu, H.; Shulgama-Morskaya, S.; Rajeev, K. G.; Manoharan, M.; Keirstead, N. D.; Maier, M. A.; Jadhav, V. Selection of GalNAc-Conjugated siRNAs with Limited off-Target-Driven Rat Hepatotoxicity. *Nat. Commun.* **2018**, 9 (1), 723.
- (12) Brown, K. M.; Nair, J. K.; Janas, M. M.; Anglero-Rodriguez, Y. I.; Dang, L. T. H.; Peng, H.; Theile, C. S.; Castellanos-Rizaldos, E.; Brown, C.; Foster, D.; Kurz, J.; Allen, J.; Maganti, R.; Li, J.; Matsuda, S.; Stricos, M.; Chickering, T.; Jung, M.; Wassarman, K.; Rollins, J.; Woods, L.; Kelin, A.; Guenther, D. C.; Mobley, M. W.; Petrusis, J.; McDougall, R.; Racie, T.; Bombardier, J.; Cha, D.; Agarwal, S.; Johnson, L.; Jiang, Y.; Lentini, S.; Gilbert, J.; Nguyen, T.; Chigas, S.; LeBlanc, S.; Poreci, U.; Kasper, A.; Rogers, A. B.; Chong, S.; Davis, W.; Sutherland, J. E.; Castoreno, A.; Milstein, S.; Schlegel, M. K.; Zlatev, I.; Charisse, K.; Keating, M.; Manoharan, M.; Fitzgerald, K.; Wu, J.-T.; Maier, M. A.; Jadhav, V. Expanding RNAi Therapeutics to Extrahepatic Tissues with Lipophilic Conjugates. *Nat. Biotechnol.* **2022**, 40 (10), 1500–1508.
- (13) Roberts, T. C.; Langer, R.; Wood, M. J. A. Advances in Oligonucleotide Drug Delivery. *Nat. Rev. Drug Discovery* **2020**, 19 (10), 673–694.
- (14) Cuellar, T. L.; Barnes, D.; Nelson, C.; Tanguay, J.; Yu, S.-F.; Wen, X.; Scales, S. J.; Gesch, J.; Davis, D.; van Brabant Smith, A.; Leake, D.; Vandlen, R.; Siebel, C. W. Systematic Evaluation of Antibody-Mediated siRNA Delivery Using an Industrial Platform of THIOMAB-siRNA Conjugates. *Nucleic Acids Res.* **2015**, 43 (2), 1189–1203.

- (15) Sela, T.; Mansø, M.; Siegel, M.; Marban-Doran, C.; Ducet, A.; Niewöhner, J.; Ravn, J.; Martin, R. E.; Sommer, A.; Lohmann, S.; Krippendorff, B.-F.; Ladefoged, M.; Indlekofer, A.; Quaiser, T.; Bueddefeld, F.; Koller, E.; Mohamed, M. Y.; Oelschlaegel, T.; Gothelf, K. V.; Hofer, K.; Schumacher, F. F. Diligent Design Enables Antibody-ASO Conjugates with Optimal Pharmacokinetic Properties. *Bioconjugate Chem.* **2023**, *34* (11), 2096–2111.
- (16) Sugo, T.; Terada, M.; Oikawa, T.; Miyata, K.; Nishimura, S.; Kenjo, E.; Ogasawara-Shimizu, M.; Makita, Y.; Imaichi, S.; Murata, S.; Otake, K.; Kikuchi, K.; Teratani, M.; Masuda, Y.; Kamei, T.; Takagahara, S.; Ikeda, S.; Ohtaki, T.; Matsumoto, H. Development of Antibody-siRNA Conjugate Targeted to Cardiac and Skeletal Muscles. *J. Controlled Release* **2016**, *237*, 1–13.
- (17) Malecova, B.; Burke, R. S.; Cochran, M.; Hood, M. D.; Johns, R.; Kovach, P. R.; Doppalapudi, V. R.; Erdogan, G.; Arias, J. D.; Darimont, B.; Miller, C. D.; Huang, H.; Geall, A.; Younis, H. S.; Levin, A. A. Targeted Tissue Delivery of RNA Therapeutics Using Antibody-Oligonucleotide Conjugates (AOCs). *Nucleic Acids Res.* **2023**, *51* (12), 5901–5910.
- (18) Cao, W.; Li, R.; Pei, X.; Chai, M.; Sun, L.; Huang, Y.; Wang, J.; Barth, S.; Yu, F.; He, H. Antibody-siRNA Conjugates (ARC): Emerging siRNA Drug Formulation. *Med. Drug Discovery* **2022**, *15*, 100128.
- (19) Chernikov, I. V.; Vlassov, V. V.; Chernolovskaya, E. L. Current Development of siRNA Bioconjugates: From Research to the Clinic. *Front. Pharmacol.* **2019**, *10*, 444.
- (20) Tai, W. Current Aspects of siRNA Bioconjugate for In Vitro and In Vivo Delivery. *Molecules* **2019**, *24* (12), 2211.
- (21) Sun, M. M. C.; Beam, K. S.; Cervený, C. G.; Hamblett, K. J.; Blackmore, R. S.; Torgov, M. Y.; Handley, F. G. M.; Ihle, N. C.; Senter, P. D.; Alley, S. C. Reduction-Alkylation Strategies for the Modification of Specific Monoclonal Antibody Disulfides. *Bioconjugate Chem.* **2005**, *16* (5), 1282–1290.
- (22) Shen, B.-Q.; Xu, K.; Liu, L.; Raab, H.; Bhakta, S.; Kenrick, M.; Parsons-Reponte, K. L.; Tien, J.; Yu, S.-F.; Mai, E.; Li, D.; Tibbitts, J.; Baudys, J.; Saad, O. M.; Scales, S. J.; McDonald, P. J.; Hass, P. E.; Eigenbrot, C.; Nguyen, T.; Solis, W. A.; Fuji, R. N.; Flagella, K. M.; Patel, D.; Spencer, S. D.; Khawli, L. A.; Ebens, A.; Wong, W. L.; Vandlen, R.; Kaur, S.; Sliwowski, M. X.; Scheller, R. H.; Polakis, P.; Junutula, J. R. Conjugation Site Modulates the in Vivo Stability and Therapeutic Activity of Antibody-Drug Conjugates. *Nat. Biotechnol.* **2012**, *30* (2), 184–189.
- (23) Zhou, Q.; Stefano, J. E.; Manning, C.; Kyazike, J.; Chen, B.; Gianolio, D. A.; Park, A.; Busch, M.; Bird, J.; Zheng, X.; Simonds-Mannes, H.; Kim, J.; Gregory, R. C.; Miller, R. J.; Brondyk, W. H.; Dhal, P. K.; Pan, C. Q. Site-Specific Antibody-Drug Conjugation through Glycoengineering. *Bioconjugate Chem.* **2014**, *25* (3), 510–520.
- (24) Li, X.; Fang, T.; Boons, G. Preparation of Well-Defined Antibody-Drug Conjugates through Glycan Remodeling and Strain-Promoted Azide-Alkyne Cycloadditions. *Angew. Chem., Int. Ed.* **2014**, *53* (28), 7179–7182.
- (25) Brun, M.-P.; Gauzy-Lazo, L. Protocols for Lysine Conjugation. In *Antibody-Drug Conjugates*; Ducry, L., Ed.; Methods in Molecular Biology; Humana Press: Totowa, NJ, 2013; Vol. 1045, pp 173–187.
- (26) Kumar, P.; Parmar, R. G.; Brown, C. R.; Willoughby, J. L. S.; Foster, D. J.; Babu, I. R.; Schofield, S.; Jadhav, V.; Charisse, K.; Nair, J. K.; Rajeev, K. G.; Maier, M. A.; Egli, M.; Manoharan, M. 5'-Morpholino Modification of the Sense Strand of an siRNA Makes It a More Effective Passenger. *Chem. Commun.* **2019**, *55* (35), 5139–5142.
- (27) Allerson, C. R.; Sioufi, N.; Jarres, R.; Prakash, T. P.; Naik, N.; Berdeja, A.; Wanders, L.; Griffey, R. H.; Swayze, E. E.; Bhat, B. Fully 2'-Modified Oligonucleotide Duplexes with Improved in Vitro Potency and Stability Compared to Unmodified Small Interfering RNA. *J. Med. Chem.* **2005**, *48* (4), 901–904.
- (28) Foster, D. J.; Brown, C. R.; Shaikh, S.; Trapp, C.; Schlegel, M. K.; Qian, K.; Sehgal, A.; Rajeev, K. G.; Jadhav, V.; Manoharan, M.; Kuchimanchi, S.; Maier, M. A.; Milstein, S. Advanced siRNA Designs Further Improve In Vivo Performance of GalNAc-siRNA Conjugates. *Mol. Ther.* **2018**, *26* (3), 708–717.
- (29) Nair, J. K.; Willoughby, J. L. S.; Chan, A.; Charisse, K.; Alam, M. R.; Wang, Q.; Hoekstra, M.; Kandasamy, P.; Kel'in, A. V.; Milstein, S.; Taneja, N.; O'Shea, J.; Shaikh, S.; Zhang, L.; Van Der Sluis, R. J.; Jung, M. E.; Akinc, A.; Hutabarat, R.; Kuchimanchi, S.; Fitzgerald, K.; Zimmermann, T.; Van Berkel, T. J. C.; Maier, M. A.; Rajeev, K. G.; Manoharan, M. Multivalent N -Acetylgalactosamine-Conjugated siRNA Localizes in Hepatocytes and Elicits Robust RNAi-Mediated Gene Silencing. *J. Am. Chem. Soc.* **2014**, *136* (49), 16958–16961.
- (30) Haraszi, R. A.; Roux, L.; Coles, A. H.; Turanov, A. A.; Alterman, J. F.; Echeverria, D.; Godinho, B. M. D. C.; Aronin, N.; Khvorova, A. 5'-Vinylphosphonate Improves Tissue Accumulation and Efficacy of Conjugated siRNAs in Vivo. *Nucleic Acids Res.* **2017**, *45* (13), 7581–7592.
- (31) Parmar, R.; Willoughby, J. L. S.; Liu, J.; Foster, D. J.; Brigham, B.; Theile, C. S.; Charisse, K.; Akinc, A.; Guidry, E.; Pei, Y.; Strapps, W.; Cancilla, M.; Stanton, M. G.; Rajeev, K. G.; Sepp-Lorenzino, L.; Manoharan, M.; Meyers, R.; Maier, M. A.; Jadhav, V. 5'-(E)-Vinylphosphonate: A Stable Phosphate Mimic Can Improve the RNAi Activity of siRNA-GalNAc Conjugates. *ChemBioChem* **2016**, *17* (11), 985–989.
- (32) Brown, C. R.; Gupta, S.; Qin, J.; Racie, T.; He, G.; Lentini, S.; Malone, R.; Yu, M.; Matsuda, S.; Shulga-Morskaya, S.; Nair, A. V.; Theile, C. S.; Schmidt, K.; Shahraz, A.; Goel, V.; Parmar, R. G.; Zlatev, I.; Schlegel, M. K.; Nair, J. K.; Jayaraman, M.; Manoharan, M.; Brown, D.; Maier, M. A.; Jadhav, V. Investigating the Pharmacodynamic Durability of GalNAc-siRNA Conjugates. *Nucleic Acids Res.* **2020**, *48* (21), 11827–11844.
- (33) Strop, P.; Liu, S.-H.; Dorywalska, M.; Delaria, K.; Dushin, R. G.; Tran, T.-T.; Ho, W.-H.; Farias, S.; Casas, M. G.; Abdiche, Y.; Zhou, D.; Chandrasekaran, R.; Samain, C.; Loo, C.; Rossi, A.; Rickert, M.; Krimm, S.; Wong, T.; Chin, S. M.; Yu, J.; Dilley, J.; Chaparro-Riggers, J.; Filzen, G. F.; O'Donnell, C. J.; Wang, F.; Myers, J. S.; Pons, J.; Shelton, D. L.; Rajpal, A. Location Matters: Site of Conjugation Modulates Stability and Pharmacokinetics of Antibody Drug Conjugates. *Chem. Biol.* **2013**, *20* (2), 161–167.
- (34) Miller, C. M.; Donner, A. J.; Blank, E. E.; Egger, A. W.; Kellar, B. M.; Østergaard, M. E.; Seth, P. P.; Harris, E. N. Stabilin-1 and Stabilin-2 Are Specific Receptors for the Cellular Internalization of Phosphorothioate-Modified Antisense Oligonucleotides (ASOs) in the Liver. *Nucleic Acids Res.* **2016**, *44* (6), 2782–2794.
- (35) Lehot, V.; Kuhn, I.; Nothisen, M.; Erb, S.; Kolodych, S.; Cianféroni, S.; Chaubet, G.; Wagner, A. Non-Specific Interactions of Antibody-Oligonucleotide Conjugates with Living Cells. *Sci. Rep.* **2021**, *11* (1), 5881.
- (36) Rocca, C.; Dennin, S.; Gu, Y.; Kim, J.; Chigas, S.; Najarian, D.; Chong, S.; Gutierrez, S.; Butler, J.; Charisse, K.; Robbie, G.; Xu, Y.; Brown, K. Evaluation of Electrophoretic Mobility Shift Assay as a Method to Determine Plasma Protein Binding of siRNA. *Bioanalysis* **2019**, *11* (21), 1927–1939.
- (37) Liang, X.; Sun, H.; Shen, W.; Crooke, S. T. Identification and Characterization of Intracellular Proteins That Bind Oligonucleotides with Phosphorothioate Linkages. *Nucleic Acids Res.* **2015**, *43* (5), 2927–2945.
- (38) Gaus, H.; Miller, C. M.; Seth, P. P.; Harris, E. N. Structural Determinants for the Interactions of Chemically Modified Nucleic Acids with the Stabilin-2 Clearance Receptor. *Biochemistry* **2018**, *57* (14), 2061–2064.
- (39) Structure–Activity Relationship of Antibody–Oligonucleotide Conjugates: Evaluating Bioconjugation Strategies for Antibody–Phosphorodiamidate Morpholino Oligomer Conjugates for Drug Development, Manuscript in Preparation.
- (40) Dosio, F.; Brusa, P.; Cattel, L. Immunotoxins and Anticancer Drug Conjugate Assemblies: The Role of the Linkage between Components. *Toxins* **2011**, *3* (7), 848–883.
- (41) Su, D.; Kozak, K. R.; Sadowsky, J.; Yu, S.-F.; Fourie-O'Donohue, A.; Nelson, C.; Vandlen, R.; Ohri, R.; Liu, L.; Ng, C.; He, J.; Davis, H.; Lau, J.; Del Rosario, G.; Cosino, E.; Cruz-Chuh, J.

D.; Ma, Y.; Zhang, D.; Darwish, M.; Cai, W.; Chen, C.; Zhou, H.; Lu, J.; Liu, Y.; Kaur, S.; Xu, K.; Pillow, T. H. Modulating Antibody-Drug Conjugate Payload Metabolism by Conjugation Site and Linker Modification. *Bioconjugate Chem.* **2018**, 29 (4), 1155–1167.

(42) Su, D.; Zhang, D. Linker Design Impacts Antibody-Drug Conjugate Pharmacokinetics and Efficacy via Modulating the Stability and Payload Release Efficiency. *Front. Pharmacol.* **2021**, 12, 687926.

(43) Kolodych, S.; Koniev, O.; Baatarkhuu, Z.; Bonnefoy, J.-Y.; Debaene, F.; Cianféroni, S.; Van Dorsselaer, A.; Wagner, A. CBTF: New Amine-to-Thiol Coupling Reagent for Preparation of Antibody Conjugates with Increased Plasma Stability. *Bioconjugate Chem.* **2015**, 26 (2), 197–200.

(44) Lyon, R. P.; Setter, J. R.; Bovee, T. D.; Doronina, S. O.; Hunter, J. H.; Anderson, M. E.; Balasubramanian, C. L.; Duniho, S. M.; Leiske, C. I.; Li, F.; Senter, P. D. Self-Hydrolyzing Maleimides Improve the Stability and Pharmacological Properties of Antibody-Drug Conjugates. *Nat. Biotechnol.* **2014**, 32 (10), 1059–1062.

(45) Meade, B. R.; Gogoi, K.; Hamil, A. S.; Palm-Apergi, C.; Berg, A. V. D.; Hagopian, J. C.; Springer, A. D.; Eguchi, A.; Kacsinta, A. D.; Dowdy, C. F.; Presente, A.; Lönn, P.; Kaulich, M.; Yoshioka, N.; Gros, E.; Cui, X.-S.; Dowdy, S. F. Efficient Delivery of RNAi Prodrugs Containing Reversible Charge-Neutralizing Phosphotriester Backbone Modifications. *Nat. Biotechnol.* **2014**, 32 (12), 1256–1261.

(46) Burke, P. J.; Hamilton, J. Z.; Jeffrey, S. C.; Hunter, J. H.; Doronina, S. O.; Okeley, N. M.; Miyamoto, J. B.; Anderson, M. E.; Stone, I. J.; Ulrich, M. L.; Simmons, J. K.; McKinney, E. E.; Senter, P. D.; Lyon, R. P. Optimization of a PEGylated Glucuronide-Monomethylauristatin E Linker for Antibody-Drug Conjugates. *Mol. Cancer Ther.* **2017**, 16 (1), 116–123.

(47) Wiggins, B.; Liu-Shin, L.; Yamaguchi, H.; Ratnaswamy, G. Characterization of Cysteine-Linked Conjugation Profiles of Immunoglobulin G1 and Immunoglobulin G2 Antibody-Drug Conjugates. *J. Pharm. Sci.* **2015**, 104 (4), 1362–1372.

(48) Prakash, T. P.; Lima, W. F.; Murray, H. M.; Li, W.; Kinberger, G. A.; Chappell, A. E.; Gaus, H.; Seth, P. P.; Bhat, B.; Crooke, S. T.; Swayze, E. E. Identification of Metabolically Stable 5'-Phosphate Analogs That Support Single-Stranded siRNA Activity. *Nucleic Acids Res.* **2015**, 43 (6), 2993–3011.

(49) Green, M. R.; Sambrook, J. Quantification of RNA by Real-Time Reverse Transcription-Polymerase Chain Reaction (RT-PCR). *Cold Spring Harb. Protoc.* **2018**, 2018 (10), pdb.prot095042.

(50) Livak, K. J.; Schmittgen, T. D. Analysis of Relative Gene Expression Data Using Real-Time Quantitative PCR and the 2- $\Delta\Delta$ CT Method. *Methods* **2001**, 25 (4), 402–408.

(51) Chen, C. Real-Time Quantification of microRNAs by Stem-Loop RT-PCR. *Nucleic Acids Res.* **2005**, 33 (20), No. e179.



CAS BIOFINDER DISCOVERY PLATFORM™

CAS BIOFINDER HELPS YOU FIND YOUR NEXT BREAKTHROUGH FASTER

Navigate pathways, targets, and
diseases with precision

Explore CAS BioFinder

